

Decentralized Learning of Stackelberg Equilibrium in General-Sum Leader-Controller Markov Games

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Abstract

We study a general-sum two-player Stackelberg Markov game under a leader-controller assumption, where both the leader and the follower aim to learn a Stackelberg equilibrium (SE) from noisy bandit feedback with unknown transition dynamics, without a central coordinator or access to best-response oracles. We introduce a notion of Stackelberg regret tailored to this setting and propose the first decentralized learning algorithm for SE in Markov games. Our algorithm combines a UCB-based learning scheme for the follower with a policy optimization method for the leader, inspired by Luo et al. [2021] and augmented with additional exploration. Notably, we establish theoretical guarantees showing that the proposed algorithm achieves sublinear Stackelberg regret under a dense transition assumption, enabling both players to converge to an approximate ε -SE. We further demonstrate the robustness of our theoretical results with numerical experiments. To the best of our knowledge, this is the first decentralized approach to Stackelberg equilibrium learning that also preserves the computational efficiency of policy optimization methods.

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1 Introduction

Multi-agent decision-making in dynamically evolving environments plays a central role in a range of applications including security games An et al. [2017], Sinha et al. [2018], online marketplaces Patel [2018], Ghasemi et al. [2020], Patel and Mishra [2025], recommendation systems Jin et al. [2018], Lin et al. [2023], autonomous control Shalev-Shwartz et al. [2016], Antonio and Maria-Dolores [2022], and human-AI collaboration Dimitrakakis et al. [2017], Nikolaidis et al. [2017], Triantafyllou et al. [2024]. In such games, one often encounters asymmetric roles between players: a leader, who commits to a policy or action sequence, and a follower, who observes this commitment and selects a best response. This asymmetry is captured by the sequential or Stackelberg model Stackelberg [1934], in which the leader anticipates the follower’s response and aims to optimize its own performance accordingly.

In *Stackelberg Markov games*, this interaction is extended to a sequential Markov decision process where the players take actions across steps, and the environment evolves in response to their actions. The leader commits to a policy at the start of the episode, and the follower responds by selecting its own policy, possibly adaptively. How these systems evolve, how algorithms can be designed to guarantee sublinear regret for both agents, and how quickly agents can converge to a Stackelberg equilibrium are all fundamental questions at the lively intersection of game theory and learning theory.

Unfortunately, these goals are generally unachievable simultaneously in Markov games without strong assumptions. When both agents’ actions influence the transition dynamics, the resulting environment is adversarial and nonstationary for each player. In particular, as the leader changes its policy over time, the transition and reward structures perceived by the follower vary in an arbitrary, policy-dependent manner, and from the leader’s perspective, the lack of reliable feedback about the follower’s behavior further hinders learning. As such, impossibility results in Markov games Foster et al. [2023], Tian et al. [2021] and recent work in *corrupted* Markov decision processes Jin et al. [2023] often preclude no-regret learning in many general-sum Stackelberg Markov games.

Despite these results, in this work, we formulate a Stackelberg Markov game with a leader-controller assumption Filar and Vrieze [2012], López et al. [2022], Zhong et al. [2023] and develop decentralized algorithms under which both individual player regret-minimization and fast Stackelberg equilibrium convergence becomes tractable. Specifically, we assume that the transition dynamics depend on only the leader’s actions, while still allowing both agents to choose arbitrary (possibly stochastic) policies. This setting models environments in which the follower’s actions are reactive or symbolic (e.g., choosing labels, outputs, or responses) but do not affect the system’s dynamics, and additionally encompasses recent lines of work assuming *myopically-rational* followers Kańska and Wiszniewska-Matyszek [2022], Chen et al. [2023], where followers only seek to maximize utility for the current time step. The key technical benefit is that the leader’s policy induces a stationary environment for the follower. This structural separation allows us to design *distributed* learning algorithms that achieve sublinear regret for both players and establish convergence to Stackelberg equilibrium.

Our contribution. In this paper, we provide the first decentralized policy optimization algorithm known for finding Stackelberg equilibrium in general-sum leader-controller Markov games with both players running no-regret algorithms. In particular, in Section 4, we develop a computationally-efficient, distributed learning, policy optimization-UCB scheme for the leader and follower and adapt it with additional exploration (Algorithm 1) that achieves sublinear *Stackelberg MDP regret* (see Section 3) on the order of $\tilde{\mathcal{O}}\left(T^{\frac{3}{4}}\right)$, under a dense transition assumption. As a direct corollary, Algorithm 1 thus ensures average-iterate convergence of the players’ policies to a Stackelberg equilibrium with rate $\mathcal{O}\left(T^{-\frac{1}{4}}\right)$. Finally, in Section 5, we demonstrate our theoretical results with numerical simulations. Our work builds upon decentralized learning in general-sum bandit games Yu et al. [2022] and the increasing popularity of policy optimization methods Schulman et al. [2017], Zhang et al. [2022], Erez et al. [2023] for finding equilibria in Markov games.

2 Related work

Learning in Markov games Markov games Shapley [1953] with simultaneous players have been extensively studied, where players face non-stationary rewards and transitions due to the other players’ actions Radanovic et al. [2019]. There has been particular attention towards decentralized settings Bai et al. [2020], Wei et al. [2021], Leonardos et al. [2021], Wang et al. [2023], Cai et al. [2023], with the goal of regret minimization and/or efficient convergence to equilibrium without exponential dependence on the number of players. In light of impossibility results on no-regret learning against adversarial opponents Tian et al. [2021], Foster et al. [2023], many works instead focus on correlated or coarse correlated equilibria. Building on multi-agent RL V-learning methods Jin et al. [2021], Song et al. [2021] and Mao and Başar [2023] develop algorithms for coarse correlated equilibrium in unknown-parameter Markov games. This finite-sample RL perspective complements the distributed-control literature on asymptotic or local convergence in repeated and state-based games Shamma and Arslan [2005], Marden and Shamma [2015]. Policy optimization methods for learning equilibria in general-sum Markov games Zhang et al. [2022], Yang and Ma [2022], Erez et al. [2023] have also attracted much interest due to their empirical efficacy Schulman et al. [2017]. Cai et al. [2024] combines V-learning with policy optimization to learn correlated equilibria at rate $\mathcal{O}(T^{-1})$, assuming access to an oracle that returns reward and transition vectors based on the policies of other players. Whether similar guarantees can be obtained in a sample-based setting where the Markov game parameters must also be learned remains open.

Stackelberg games In a more closely related vein, sequential or Stackelberg games have been studied in many forms, including Stackelberg security games Conitzer and Sandholm [2006], Tian et al. [2021], dynamic pricing Kleinberg and Leighton [2003], online contract design Zhu et al. [2022], and strategic learning Dong et al. [2018]. In zero-sum two-player bandit games, Stackelberg equilibria coincide with Nash equilibria under convex-concave

assumptions by von Neumann’s minmax theorem von Neumann [1959], and average-iterate convergence follows in many regret-minimization settings Freund and Schapire [1999], Cai et al. [2025]. General-sum Stackelberg games, especially Markov games, are substantially more complex.

An important distinction in this literature is between centralized and decentralized settings. In centralized Stackelberg games Letchford et al. [2009], Blum et al. [2014], Peng et al. [2019], a central authority or oracle can coordinate players’ decisions, making equilibrium computation and learning more tractable. In the bandit setting, Bai et al. [2021] study repeated Stackelberg games where a central controller determines joint actions to estimate empirical means. Gerstgrasser and Parkes [2023] and Zhong et al. [2023] both study Stackelberg equilibrium learning in Markov games, with the latter also assuming a leader-controller Markov game, but both rely on an oracle or central coordinator that computes policies constituting approximate best-response equilibria.

In contrast, this work, to the best of our knowledge, presents the first algorithm for decentralized Stackelberg equilibrium learning in general-sum Stackelberg Markov games. Rather than relying on best-response oracles Zhao et al. [2023], players learn by running individual online learning algorithms across rounds Cai et al. [2025]. This builds on work in the bandit setting: Goktas et al. [2022] and Sun et al. [2023] study the simpler zero-sum case, while Yu et al. [2022] considers individual regret minimization in general-sum games, showing that a leader using exploration-modified Exp3 Lattimore and Szepesvári [2020] together with a follower using UCB converges to an approximate Stackelberg equilibrium at rate $\mathcal{O}(T^{-1/3})$. Lauffer et al. [2023] and Balcan et al. [2015] instead study settings where the follower best-responds to the leader’s randomized policy, rather than the realized action. Yu and Chen [2024] consider alternative information structures where the follower is either myopic or can observe the leader’s rewards, while Kao et al. [2022] study a cooperative decentralized two-player sequential MDP in which both players using UCB ensure no-regret learning. This work is also related to federated multiplayer reinforcement learning Jiang

et al. [2013], Balcan et al. [2015], Letchford et al. [2009], Marecki et al. [2012], where privacy, security, and computational efficiency Tambe [2011] are also central concerns.

3 Preliminary and problem formulation

3.1 Stackelberg Markov game

We formulate a tabular episodic two-player sequential MDP ("Stackelberg Markov game") with bandit feedback given by $(\mathcal{S}, \mathcal{A}, \mathcal{K}, H, P, l, l_f)$ played by one leader and one follower across T episodes (the "horizon"), where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action set of the leader, $\mathcal{K}$ is the action set of the follower, and H is the number of steps taken in each episode. We denote $|\mathcal{S}| = S$, $|\mathcal{A}| = A$, and $|\mathcal{K}| = K$. Furthermore, we assume that the states admit a layered structure where $\mathcal{S}$ can be partitioned into $\mathcal{S}_0, \mathcal{S}_1, \dots, \mathcal{S}_H$ and where $\mathcal{S}_0$ contains only the initial state s_0 and $\mathcal{S}_H$ contains only the final state s_H . The MDP comes equipped with a loss function for the leader $l : \mathcal{S} \times \mathcal{A} \times \mathcal{K} \rightarrow [0, 1]$ and a loss function for the follower, $l^f : \mathcal{S} \times \mathcal{A} \times \mathcal{K} \rightarrow [0, 1]$ which both depend on the current state and both players' actions. However, the leader and follower, after visiting a state and taking their respective actions, observe "noisy" losses, $\ell \in [0, 1]$ and $\ell^f \in [0, 1]$ respectively, such that $\mathbb{E}[\ell(s, a, k)] = l(s, a, k)$ and $\mathbb{E}[\ell^f(s, a, k)] = l^f(s, a, k)$ for any $(s, a, k) \in \mathcal{S} \times \mathcal{A} \times \mathcal{K}$. In this work, we assume a *dense, leader-controller* Markov game Filar and Vrieze [2012], López et al. [2022], Zhong et al. [2023]. Formally, we make the following assumptions about the transitions P :

Assumption 1 *Let P be the transition function such that for some $s \in \mathcal{S}_h$, $h = 0, \dots, H-1$, and some $a \in \mathcal{A}$, $P(\cdot|s, a)$ gives the distribution over the states in $\mathcal{S}_{h+1}$ given that the leader took action a in state s on step h .*

Notably, Assumption 1 also implies the follower is "myopic" and only seeks to minimize their instantaneous loss for any h -step. Such an assumption is well-motivated in Stackelberg contexts; for example, a first-mover firm's decision on output can not only control the profit

they receive, but also deter the entry of competing firms, affecting the state of the industry in the next time step Shum [2023], or a government can act as a leader whose price-setting can affect firm entry. Myopic responses also cover allocation, routing, and local response decisions.

For each episode t , the leader decides on a policy $\pi_t : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ where $\Delta(\mathcal{A})$ denotes the set of distributions over $\mathcal{A}$ and $\pi_t(a|s)$ gives the probability of the leader choosing action a in state s . The follower also decides on a policy $\pi_t^f : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{K})$, where $\Delta(\mathcal{K})$ denotes the set of distributions over $\mathcal{K}$ and $\pi_t^f(k|s, a)$ gives the probability of the follower choosing action k given the players arrive in state s and the leader has taken action a . We summarize the interaction protocol between the two players and the MDP for an episode t as follows:

- Leader chooses π_t and follower chooses π_t^f . They start at state s_0 .
- For every $h \in \{0, 1, \dots, H-1\}$ and the respective state visited s_h ,
 1. Leader takes action a_h sampled from $\pi_t(\cdot|s_h)$.
 2. Follower observes the leader's a_h and takes action k_h sampled from $\pi_t^f(\cdot|s_h, a_h)$.
 3. Leader observes and suffers *noisy* loss $\ell(s_h, a_h, k_h)$ and follower observes and suffers *noisy* loss $\ell_f(s_h, a_h, k_h)$.
 4. The players move to s_{h+1} sampled from $P(\cdot|s_h, a_h)$.
- Once s_H is reached, the episode terminates, and the leader and follower observe the feedback and update their policies accordingly.

We emphasize how the players seek to minimize their own incurred losses and do not observe each other's losses (relevant for equilibrium in Section 3.3). "Decentralized" here refers to how both agents learn from their own bandit feedback without a centralized planner or best-response oracle. The follower observes the leader action before responding, whereas the leader observes only its own noisy loss and does not observe the follower action. With this in mind, the leader's observed losses are adversarial, as they depend on the follower's

(potentially arbitrary) action. We utilize this intuition in our algorithm construction and subsequent analysis.

3.2 Value functions

Given a leader loss function ℓ , leader policy π , and follower policy π^f , we can define the leader's expected sum of losses for the remainder of an episode starting from state $s_{h'}$ in some layer h' , $h' = 0, 1, \dots, H - 1$, as:

$$V^{\pi, \pi^f}(s_{h'}; \ell) := \mathbb{E}_{a_h \sim \pi(\cdot | s_h), k_h \sim \pi^f(\cdot | a_h, s_h), s_{h+1} \sim P(\cdot | s_h, a_h)} \left[\sum_{h=h'}^{H-1} \ell(s_h, a_h, k_h) \right] \quad (1)$$

where V is called the leader's *state value function* or V -value function, given the leader follows policy π and the follower follows policy π^f . We can also define the leader's value function using the Bellman equations that involve the leader's *state-action value function*, or Q -value function, defined for some layer $h = 0, 1, \dots, H - 1$ as:

$$Q^{\pi, \pi^f}(s_h, a_h; \ell) := \mathbb{E}_{k_h \sim \pi^f(\cdot | s_h, a_h)} [\ell(s_h, a_h, k_h)] + \mathbb{E}_{s_{h+1} \sim P(\cdot | s_h, a_h)} [V^{\pi, \pi^f}(s_{h+1}; \ell)]$$

$$V^{\pi, \pi^f}(s_h; \ell) = \mathbb{E}_{a_h \sim \pi(\cdot | s_h)} [Q^{\pi, \pi^f}(s_h, a_h; \ell)]$$

where we initialize $V^{\pi, \pi^f}(s_H) := 0$. We remark that since the follower is myopically-rational, the problem from the follower's perspective is that of a stochastic multi-armed bandit with SA unique environments, and hence, optimal follower policies will be deterministic. In cases where the follower may have more than one best response to any state-leader action, Stackelberg equilibrium becomes more ambiguous, and the literature Conitzer and Sandholm [2006], Breton et al. [1988], López et al. [2022] often assumes the follower chooses the response which is better (optimistic) or worse off (pessimistic) for the leader. In this work, we remove this ambiguity by making the standard game theory and online learning assumption Auer et al. [2002], Aussel and Svensson [2020]:

Assumption 2 *There exists a unique best-response follower action in each state-leader action environment.*

We define the follower’s optimal or *best-response policy*, $\pi_\star^f$, to be that for which for each $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$\pi_\star^f(s, a) = \arg \min_{k \in \mathcal{K}} l^f(s, a, k). \quad (2)$$

We define $\varepsilon := \min_{s \in \mathcal{S}, a \in \mathcal{A}, k \neq \pi_\star^f(s, a)} \Delta_{s, a, k}$, where $\Delta_{s, a, k} = l^f(s, a, k) - l^f(s, a, \pi_\star^f(s, a))$ to be the minimum suboptimality gap for the follower. We assume that the self-interested follower’s objective is to learn $\pi_\star^f$. Thus, Assumption 2 provides a stable and statistically identifiable target, as the positive gap permits UCB to identify the unique response from finitely many noisy observations. When follower actions tie, one must instead specify an optimistic, pessimistic, or set-valued Stackelberg benchmark and modify the follower-learning analysis accordingly.

3.3 Stackelberg equilibrium and regret

We study equilibrium in Stackelberg Markov games where both the leader and the follower are best-responding to each other’s policy, so that neither can improve their outcome by unilaterally deviating. In particular, we approach Stackelberg equilibrium (SE) from a learning perspective Bai et al. [2021], Yu et al. [2022], Zhong et al. [2023], where the players attempt to learn a SE policy profile within some degree of accuracy across shorter horizons T . We define for the Markov game:

Definition 1 (Stackelberg equilibrium) *For a Stackelberg Markov game satisfying Assumptions 1 and 2, a policy pair $(\pi, \pi_\star^f)$ is an ϵ -**approximate ex-ante Stackelberg equilibrium** if*

$$V^{\pi, \pi_\star^f}(s_0) \leq V^{\pi_\star, \pi_\star^f}(s_0) + \epsilon,$$

*and is an ϵ -**approximate state-wise Stackelberg equilibrium** if the same inequality holds for every $s \in \mathcal{S}$.*

The learned follower policy π_t^f is time-varying, whereas $\pi_\star^f$ is the fixed best-response mapping of the underlying stationary game and the target used in the Stackelberg benchmark. In this paper, we measure approximation error ε through the learned leader policy after pairing it with $\pi_\star^f$.

In this work, we seek to develop the first *decentralized* algorithms that provably guarantee Stackelberg equilibrium convergence in Stackelberg Markov games. To this end, we define a new objective: *MDP Stackelberg-regret*, given by:

$$R_{\text{Stack}}(T) := \sum_{t=1}^T V^{\tilde{\pi}_t, \pi_\star^f}(s_0; l) - \min_{\pi} \sum_{t=1}^T V^{\pi, \pi_\star^f}(s_0; l). \quad (3)$$

The second term of Equation 3 is the total leader loss in SE, and so we see algorithms whose MDP Stackelberg regret scale sub-linearly with T converge to a SE. Note that in the first term of Equation 3, the follower’s best-response policy is fixed as well, which differentiates Stackelberg regret analysis from that of a traditional single-agent adversarial MDP, where the first term would instead be the expected incurred losses of the leader during play (with the follower playing $\{\pi_t^f\}_{1 \leq t \leq T}$). Stackelberg regret thus intuitively incorporates the ”learning capacity” of the regret-minimizing follower; indeed, if $\pi_t^f \neq \pi_\star^f$ for many t , this may prevent the leader from adequately learning $\pi_\star$ themselves, leading to increased *MDP Stackelberg regret*.

4 Policy optimization algorithm for SE

In this section, we introduce Algorithm 1, a decentralized policy optimization algorithm for Stackelberg equilibrium with loss-minimizing players. The crux of Algorithm 1 lies in the rather-intuitive combination of no-regret algorithms: an Upper Confidence Bound (UCB)-based algorithm for the follower with a no-regret adversarial MDP policy optimization scheme inspired by Luo et al. [2021] for the leader, which lends itself to high computational efficiency and strong empirical performance. The pseudocode is given in Algorithm 1, with

supporting details and formulas deferred to Appendix A.

Algorithm 1 Policy Optimization with α Exploration and Upper Confidence Bound

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1: Input: confidence parameter  $\delta \in (0, 1)$ , exploration parameter  $\alpha \in (0, 1)$ 
2: Leader initialization:
3:  $\eta \leftarrow \min\{1/(24H^3), 1/\sqrt{HSAT}\}$ ,  $\gamma \leftarrow 2\eta H$ 
4: Set epoch  $k \leftarrow 1$  and  $\mathcal{P}_1$  to be the set of all transition functions
5: Set epoch counters for all  $(s, a, s') \in \mathcal{S} \times \mathcal{A} \times \mathcal{S}$ :
6:    $N_0(s, a), N_1(s, a) \leftarrow 0$ ;  $N_0(s, a, s'), N_1(s, a, s') \leftarrow 0$ 
7: Follower initialization:
8: for all  $(s, a, k)$  do
9:    $n_{(s,a)}(0) \leftarrow 0$ ;  $n_k^f(0) \leftarrow 0$ ;  $\hat{l}_{s,a,k}^f(0) \leftarrow 0$ 
10: end for
11: for  $t = 1, 2, \dots, T$  do
12:   Leader computes  $\pi_t$  such that  $\pi_t(a|s) \propto \exp\left(-\eta \sum_{\tau=1}^{t-1} \left(\hat{Q}_\tau(s, a) - B_\tau(s, a)\right)\right)$ 
13:   Leader sets, for all  $(s, a) \in \mathcal{S} \times \mathcal{A}$ ,  $\tilde{\pi}_t(a|s) = (1 - \alpha)\pi_t(a|s) + \alpha/A$ 
14:   Follower computes UCBs for all  $(s, a, k)$  using Eq. (4)
15:   Follower chooses  $\pi_t^f$  such that  $\pi_t^f(s, a) \in \arg \min_k \text{UCB}_{(s,a)}(k)$  for all  $(s, a) \in \mathcal{S} \times \mathcal{A}$ 
16:   Execute policies  $(\tilde{\pi}_t, \pi_t^f)$  and receive trajectories
17:   for all  $(s, a)$  do
18:     Compute  $\hat{Q}_t(s, a)$  using Eq. (6) and compute  $B_t(s, a)$  using Eqs. (7) and (8)
19:   end for
20:   for  $h = 0, 1, \dots, H - 1$  do
21:      $N_k(s_{t,h}, a_{t,h}) \stackrel{+}{\leftarrow} 1$ ;  $N_k(s_{t,h}, a_{t,h}, s_{t,h+1}) \stackrel{+}{\leftarrow} 1$ 
22:   end for
23:   if  $\exists (s, a)$  such that  $N_k(s, a) \geq \max\{1, 2N_{k-1}(s, a)\}$  then
24:      $k \stackrel{+}{\leftarrow} 1$ ;  $N_k \leftarrow N_{k-1}$ 
25:     Compute  $\mathcal{P}_k$  using Eq. (9) in Appendix A
26:   end if
27:   for all  $(s, a, k)$  do
28:     Update  $n_{(s,a)}(t)$  and update  $n_k^f(n_{(s,a)}(t))$ 
29:     Compute  $\hat{l}_{s,a,k}^f(t)$  using Eq. (5)
30:   end for
31: end for

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The policy optimization-based Algorithm 1 of Luo et al. [2021] achieves $\tilde{\mathcal{O}}(\sqrt{T})$ regret in adversarial MDPs by computing stochastic policies using exponential weight updates on Q -value estimates, $\hat{Q}_t$, for each state-action pair each round. These estimates are built using an estimated model where optimistic state-action occupancy measures under some policy π_t ($\bar{q}_t(s, a)$) are computed by taking the maximum over transition functions within a confidence

set (see COMP-UOB, Algorithm 3 of Jin et al. [2020]). This confidence set is carefully constructed around an empirical estimate of the transition to contain the true transition with high probability and is updated when the number of visits to some state-leader action pair doubles. The period between two updates is called an epoch, and the transition confidence set in epoch k is denoted $\mathcal{P}_k$. We remark that because of Assumption 1, the leader is able to estimate the *transitions* from their own actions taken in this manner. Furthermore, to address the global exploration limitations of policy optimization, dilated bonuses B_t that utilize the high-probability occupancy bounds are subtracted from the Q -estimates to encourage exploration of rarely visited states and actions.

On the myopically-rational follower's side, learning $\pi_\star^f$ amounts to identifying the unique best-response $\pi_\star^f(s, a) \in \mathcal{K}$ for each state-leader action pair (s, a) . The follower can thus run SA UCB algorithms, treating each (s, a) as an independent environment. For a given round t and for each (s, a) visited, the follower chooses $\pi_t^f(s, a) \in \arg \min_k \text{UCB}_{(s,a)}(k)$, where for all (s, a, k) , $\text{UCB}_{(s,a)}(k)$ is computed by:

$$\text{UCB}_{(s,a)}(k) := \widehat{l}_{s,a,k}^f(t-1) - \sqrt{\frac{6 \log n_{(s,a)}(t-1)}{n_k^f(n_{(s,a)}(t-1))}} \quad (4)$$

where $n_{(s,a)}(t-1)$ is the number of times state-leader action (s, a) is visited up to time t , $n_k^f(x)$ is the number of times the follower takes response k across x instances of some state-leader action (s, a) , and $\widehat{l}_{s,a,k}^f(t-1)$ is the empirical mean loss estimate up to round t from taking k in (s, a) updated as follows:

$$\widehat{l}_{s,a,k}^f(t) = \frac{\widehat{l}_{s,a,k}^f(t-1) n_k^f(n_{(s,a)}(t-1)) + \sum_h \ell^f(s_{t,h}, a_{t,h}, k_{t,h}) \mathbb{I}[(s, a, k) = (s_{t,h}, a_{t,h}, k_{t,h})]}{n_k^f(n_{(s,a)}(t))}. \quad (5)$$

The UCB is set to $-\infty$ if $n_k^f(n_{(s,a)}(t-1)) = 0$. We remark that because of the partitioned state-space, any (s, a) can be visited a maximum of one time per episode.

4.1 α -Exploration, estimators, and bonuses

A key novel contribution needed to adapt this framework for *MDP Stackelberg regret* minimization is using the policy $\tilde{\pi}_t$, which incorporates a fixed explicit exploration parameter α into the leader's policy taken from exponential weights, π_t . This ensures that in any state, the probability of selecting any action a is at least α/A . This extends from the exploration mechanism in Yu et al. [2022] for the bandit setting, ensuring that the leader plays all actions often enough for the follower to reliably learn its best response.

We remark that the Stackelberg regret benchmark has a learning-to-commit interpretation. Stackelberg equilibrium is the full-information commitment solution in which the leader is evaluated against the follower's true best response; it is not an equilibrium over the players' choices of learning algorithms. Exploration therefore lets the leader identify which actions remain attractive after the follower best responds, rather than helping the follower by itself. Strategic teaching or manipulation of a transient learner would require a richer dynamic game over learning rules and is outside our scope.

Choosing α to be on the order of $\mathcal{O}(T^{-1/4})$ balances the sampling needs of the follower with regret-minimization of the leader, guaranteeing sublinear MDP Stackelberg regret as highlighted in Theorem 1. Note that such an explicit exploration technique can only be run on the level of the action set, not of the states. For the theoretical analysis, we make the following additional *density* assumption on P :

Assumption 3 *We let $\beta := \min\{P(s_{h+1}|s_h, a) : (s_h, a, s_{h+1}) \in \mathcal{S}_h \times \mathcal{A} \times \mathcal{S}_{h+1}, h \in \{0, 1, \dots, H-1\}\}$ be the minimum transition probability and assume $\beta > 0$.*

Assumption 3 states that each state in the Markov game can be visited by the players with a nonzero probability from any state in the previous step, i.e., the transition matrices are dense. Such reachability assumptions naturally arise in many applications; for example, exogenous environmental or policy shocks, macroeconomic fluctuations, or general stochastic perturbations. Without Assumption 3, a state could hypothetically only be reached via one

trajectory, resulting in, at worst, an exponential in H dependency on the probability that some (s, a) is reached. It is appropriate in finite episodic systems with exogenous transition noise, random resets, or trembling dynamics, or after restricting attention to a uniformly reachable subset. It is less suitable for sparse or nearly deterministic routing, inventory, scheduling, or networked systems. In Section 5, we test the robustness of our algorithm without Assumption 3.

Moreover, since leader loss feedback now comes from $\tilde{\pi}_t$, we diverge from Luo et al. [2021] by constructing Q -value estimates for $Q_t^{\tilde{\pi}_t}$. Formally, let $\{s_{t,h}, a_{t,h}, k_{t,h}, \ell(s_{t,h}, a_{t,h}, k_{t,h})\}_{h=0}^H$ be the trajectory received by the leader on episode t taking policy $\tilde{\pi}_t$ and let k be the current epoch. For $h \in \{0, 1, \dots, H\}$, $(s, a) \in \mathcal{S}_h \times \mathcal{A}$, we compute:

$$\hat{Q}_t(s, a) := \frac{L_{t,h}}{\bar{q}_t(s, a) + \gamma} \mathbb{I}_t(s, a) \quad (6)$$

where $L_{t,h} := \sum_{i=h}^H \ell(s_{i,h}, a_{i,h}, k_{i,h})$, $\bar{q}_t(s, a) := \max_{\hat{P} \in \mathcal{P}_k} q^{\hat{P}, \tilde{\pi}_t}(s, a)$, and $\mathbb{I}_t(s, a) := \mathbb{I}[s_{t,h} = s, a_{t,h} = a]$. Importantly, the upper occupancy bound $\bar{q}_t(s, a)$ is computed under the *modified* policy $\tilde{\pi}_t$ to make $\hat{Q}$ an appropriate optimistic under-estimator for $Q^{\tilde{\pi}_t}$.

Our last algorithmic novelty is making an adjustment to the dilated bonus construction in Luo et al. [2021]. Formally, let k be the current epoch and we compute for all $(s, a) \in \mathcal{S} \times \mathcal{A}$:

$$b_t(s) = \mathbb{E}_{a \sim \pi_t(\cdot|s)} \left[\frac{3\gamma H + H(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} \right] \quad (7)$$

$$B_t(s, a) = b_t(s) + \left(1 + \frac{1}{H}\right) \max_{\hat{P} \in \mathcal{P}_k} \mathbb{E}_{s' \sim \hat{P}(\cdot|s, a)} \mathbb{E}_{a' \sim \pi_t(\cdot|s')} [B_t(s', a')] \quad (8)$$

where $\underline{q}_t(s, a) = \min_{\hat{P} \in \mathcal{P}_k} q^{\hat{P}, \tilde{\pi}_t}(s, a)$, $B_t(s_H, a) = 0$ for all a . Notice how the upper and lower occupancy bounds are computed with respect to the actual policy taken, $\tilde{\pi}_t$, but the *expectation over actions in the states* is taken with respect to the *unmodified* policy drawn from exponential weights, π_t . This subtle change is crucial for bounding the regret of the leader's learning process, ensuring that we can use the recursion in Lemma 3.1 of Luo et al.

[2021] to control the regret incurred from bonuses. We now state the main result for Algorithm 1.

Theorem 1 *In the T -episodic Stackelberg Markov game setting (Section 3), if the leader and follower apply Algorithm 1 and set $\alpha = \mathcal{O}(H^{-1}S^{\frac{1}{2}}A(\log A)^{\frac{1}{3}}T^{-\frac{1}{4}})$, with probability at least $1 - \mathcal{O}(\delta)$, the MDP Stackelberg regret is bounded as:*

$$R_{\text{Stack}}(T) \leq \tilde{\mathcal{O}} \left(\frac{H^2 S A K T^{\frac{1}{4}}}{\beta \varepsilon^2} + H^4 + H^3 S A^{\frac{1}{2}} T^{\frac{3}{4}} + H S^{\frac{1}{2}} A T^{\frac{3}{4}} \right)$$

where $\tilde{\mathcal{O}}$ hides logarithmic dependencies.

Notably, Theorem 1 shows that Algorithm 1 establishes the first sublinear Stackelberg regret guarantee for decentralized Stackelberg Markov game setting on the order of $\mathcal{O}(T^{3/4})$ with high probability, extending current guarantees in the decentralized bandit setting Yu et al. [2022] while preserving high computational efficiency with policy optimization methods. Full details on computing the transition confidence set $\mathcal{P}_k$, occupancy bounds, and other algorithmic details can be found in Appendix A. The full proof of Theorem 1 is in Appendix C, and the key details of the analysis are highlighted in Section 4.2.

Algorithm 1 yields two distinct output policies for implementing SE with $\mathcal{O}(T^{-\frac{1}{4}})$ approximation error. For the standard ex-ante SE guarantee, we implement using the following random policy: sample $J \sim \text{Unif}([T])$ once at the beginning of an episode and execute $\tilde{\pi}_J$ throughout that episode. We denote this episode-level mixture by $\hat{\pi}$.

Theorem 2 (Ex-ante equilibrium) *Under the conditions of Theorem 1, with probability at least $1 - \mathcal{O}(\delta)$, the pair $(\hat{\pi}, \pi_f^*)$ is an ϵ_T^{ex} -approximate ex-ante Stackelberg equilibrium, where*

$$\epsilon_T^{\text{ex}} = \frac{R_{\text{Stack}}(T)}{T} = \tilde{\mathcal{O}} \left(\frac{H^2 S A K T^{-\frac{3}{4}}}{\beta \varepsilon^2} + H^4 T^{-1} + H^3 S A^{\frac{1}{2}} T^{-\frac{1}{4}} + H S^{\frac{1}{2}} A T^{-\frac{1}{4}} \right).$$

Assumption 3 additionally allows Algorithm 1 to obtain the stronger state-wise guarantee.

To implement this, we instead use the pointwise Markov average,

$$\bar{\pi}(a \mid s) := \frac{1}{T} \sum_{t=1}^T \tilde{\pi}_t(a \mid s).$$

Theorem 3 (State-wise equilibrium) *Under the conditions of Theorem 1, including Assumption 3, with probability at least $1 - \mathcal{O}(\delta)$, the pair $(\bar{\pi}, \pi_f^*)$ is an ϵ_T^{sw} -approximate state-wise Stackelberg equilibrium, where*

$$\epsilon_T^{\text{sw}} = \frac{H}{\beta} \frac{R_{\text{Stack}}(T)}{T} = \tilde{\mathcal{O}} \left(\frac{1}{\beta} \left(\frac{H^3 S A K T^{-\frac{3}{4}}}{\beta \varepsilon^2} + H^5 T^{-1} + H^4 S A^{\frac{1}{2}} T^{-\frac{1}{4}} + H^2 S^{\frac{1}{2}} A T^{-\frac{1}{4}} \right) \right).$$

The proofs of Theorems 2 and 3 are given separately in Appendices D and E. Theorem 2 gives the standard, assumption-light ex-ante equilibrium policy guarantee, yielding an approximation error equal to the average Stackelberg regret. Theorem 3 states that the approximation error for the state-wise SE policy is equal to the average Stackelberg regret, with an additional H/β factor incurred from passing the small regret at s_0 to all states in the Markov game using Assumption 3.

4.2 Proof sketch

We sketch a high-level overview of the proof of our main result, Theorem 1, here and defer the details to Appendix C. We first decompose $R_{\text{Stack}}(T)$ into the terms:

$$\begin{aligned} R_{\text{Stack}}(T) &= \sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_*^f}(s_0) - V^{\pi_*, \pi_*^f}(s_0) \right) = \underbrace{\sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_*^f}(s_0) - V^{\tilde{\pi}_t, \pi_t^f}(s_0) \right)}_{\text{(Term I)}} \\ &\quad + \underbrace{\sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_t^f}(s_0) - V^{\pi_*, \pi_t^f}(s_0) \right)}_{\text{(Term II)}} + \underbrace{\sum_{t=1}^T \left(V^{\pi_*, \pi_t^f}(s_0) - V^{\pi_*, \pi_*^f}(s_0) \right)}_{\text{(Term III)}}. \end{aligned}$$

Terms I and III capture the learning of the follower, and are small by 1) shrinking UCB intervals and 2) the explicit $\alpha = \mathcal{O}(T^{-\frac{1}{4}})$ exploration parameter, which allows sufficient sampling. Lemmas 7 and 10 show these terms are bounded well with high probability. Term II encapsulates the leader's learning. We first suppress the value functions' dependence on the follower's policies (π_t^f) by treating the leader's losses as adversarial. We then use the performance difference lemma Kakade and Langford [2002], Luo et al. [2021] to bound Term II by:

$$\text{Term II} \leq \sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \sum_{a \in \mathcal{A}} (\tilde{\pi}_t(a|s) - \pi_*(a|s)) Q_t^{\tilde{\pi}_t}(s, a)$$

where $q^\pi(s)$ is the probability of visiting state s following policy π and Q_t^π is the (hypothetical) adversarial Q -value function that has suppressed the follower's policies. Next, to control regret from bonuses, we use Lemma 3.1 from Luo et al. Luo et al. [2021] and perform the decomposition $\sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \langle \pi_t(\cdot|s) - \pi^*(\cdot|s), Q_t^{\tilde{\pi}_t}(s, \cdot) - B_t(s, \cdot) \rangle = \text{BIAS-1} + \text{BIAS-2} + \text{REG-TERM}$, where

- $\text{BIAS-1} = \sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \langle \pi_t(\cdot|s), Q_t^{\tilde{\pi}_t}(s, \cdot) - \hat{Q}_t(s, \cdot) \rangle$ and $\text{BIAS-2} = \sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \langle \pi^*(\cdot|s), \hat{Q}_t(s, \cdot) - B_t(s, \cdot) \rangle$ are small due to the fact that we constructed $\hat{Q}_t$ to be good optimistic estimators of $Q_t^{\tilde{\pi}_t}$, and
- $\text{REG-TERM} = \sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \langle \pi_t(\cdot|s) - \pi^*(\cdot|s), \hat{Q}_t(s, \cdot) - B_t(s, \cdot) \rangle$ is small since the unmodified π_t is drawn from the exponential weights algorithm with $\hat{Q}_t - B_t$ as loss vectors.

Finally, in Lemma 9, we resolve policy mismatches in occupancy measures by showing that for any s , $q^{\hat{P}_t, \pi_t}(s)$ is upper bounded by $q^{\hat{P}_t, \tilde{\pi}_t}(s)$ times a factor of just $(1 - \alpha)^{-H}$, which is small for large T with our choice of α . Lemma 8 shows that by balancing our choice of α and η , we have that with probability at least $1 - \mathcal{O}(\delta)$,

$$\text{Term II} \leq \tilde{O} \left(H^4 + H^3 S A^{\frac{1}{2}} T^{\frac{3}{4}} + H S^{\frac{1}{2}} A T^{\frac{3}{4}} \right).$$

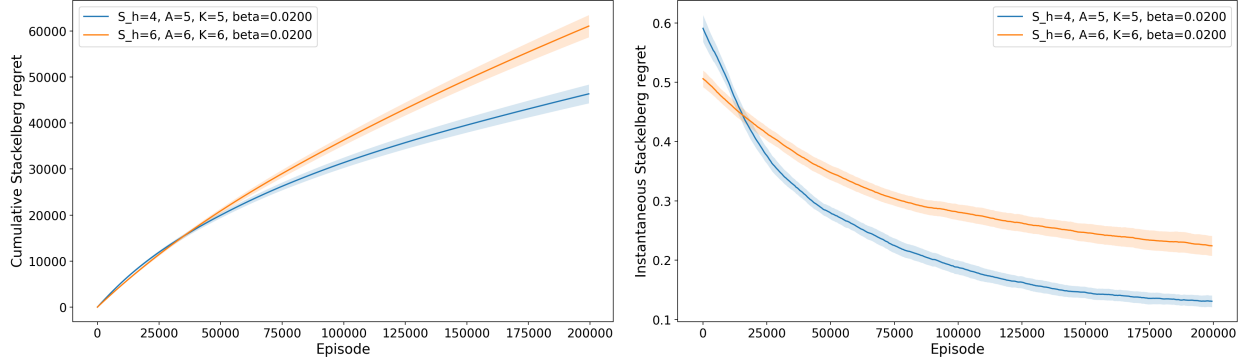


Figure 1: Mean Stackelberg regret and convergence to SE for Algorithm 1, $\beta = 0.02$

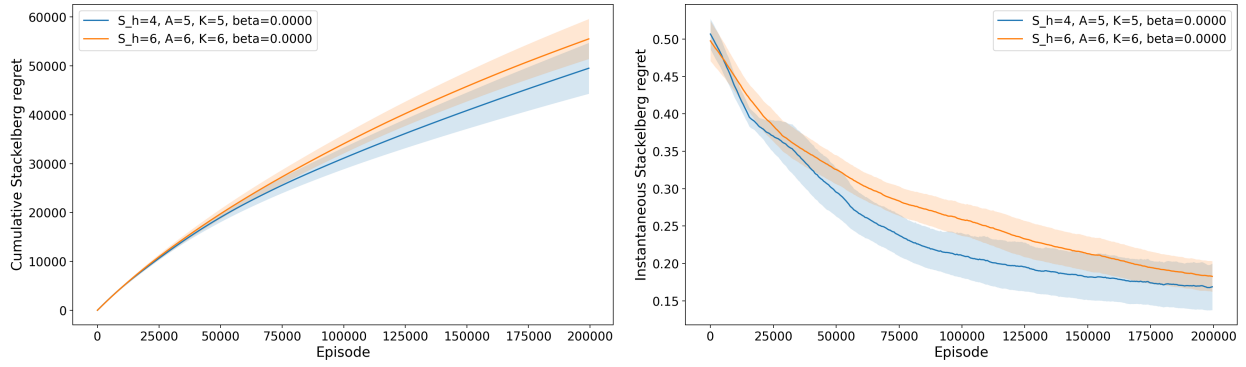


Figure 2: Mean Stackelberg regret and convergence to SE for Algorithm 1, $\beta = 0$

Chaining terms I, II, and III gives Theorem 1.

5 Numerical experiments

To complement the analysis above, we numerically evaluate Algorithm 1 over a range of state and action sizes with noisy Bernoulli bandit losses. In each experiment, we run $T = 200,000$ episodes on a finite-horizon game with horizon $H = 3$, and compare runs across $(S_h, A, K) \in \{(4, 5, 5), (6, 6, 6)\}$ ($S_h = |\mathcal{S}_1| = |\mathcal{S}_2|$) to illustrate how the regret behavior changes with problem size. We further separate the experiments into two transition regimes. In the first, the transition kernel is constructed so that $\beta = 0.02$, matching Assumption 3 used in the theory. In the second, the environment randomly generates transition dynamics, violating the $\beta > 0$ assumption. Figure 1 shows how Algorithm 1 achieves sublinear

growth of Stackelberg regret and convergence to the SE (instantaneous Stackelberg regret) for $\beta > 0$ environments, verifying claims in Theorems 1 and 2. Bands show ± 1 standard error across 5 independent random seeds, where each seed affects both environment generation and the stochasticity of the learning dynamics (action sampling, transition sampling, and noisy feedback). Figure 2 shows that Algorithm 1 is in fact robust to more sparse transition environments without $\beta > 0$, still achieving sublinear-like growth across runs. However, larger bands in the $\beta = 0$ setting indicate greater run-to-run variability, consistent with the idea that state-action visitation becomes less uniform making SE learning more difficult.

6 Conclusion and Future Directions

In this paper, we proposed the first provably efficient decentralized policy optimization algorithm for SE learning in general-sum leader-controller Markov games. These results reinforce those in Freund & Schapire Freund and Schapire [1999] and partially address the open question posed in Zhong et al. Zhong et al. [2023] of learning SE with greater computational efficiency and no oracle. This framework connects to broader multi-agent decision-making problems, such as security, marketplaces, and human–AI collaboration, where efficient, often decentralized, learning is critical. We leave various directions to explore. In practice, system evolution may meaningfully depend on followers’ actions or even other simultaneous leaders. Our experimental results also hinted at the effectiveness of Algorithm 1 on sparse environments, suggesting our theoretical bounds can be improved. Thus, we ask the challenging question as to whether decentralized algorithms can provably learn (correlated) SE in Markov games with more general transition dynamics depending on all players. A further open direction is exploring whether the $\tilde{O}(T^{3/4})$ Stackelberg regret rate is information-theoretically optimal, potentially exploiting the coupled tradeoff between leader exploration and follower best-response identification. Finally, we ask if our results can also be extended to the trickier *last-iterate* convergence Mertikopoulos et al. [2018], Bailey and

Piliouras [2018], Cai et al. [2023, 2025].

Impact statement This paper is primarily theoretical. While such results may help inform future work on strategic multi-agent systems with limited feedback, the present contribution is algorithmic and analytical rather than deployment-oriented. Accordingly, the broader implications depend on future work that addresses issues such as robustness, fairness, and domain-specific safeguards.

References

- Bo An, Milind Tambe, and Arunesh Sinha. Stackelberg security games (ssg) basics and application overview. *Improving Homeland Security Decisions*, 2:485, 2017.
- Guillen-Perez Antonio and Cano Maria-Dolores. Multi-agent deep reinforcement learning to manage connected autonomous vehicles at tomorrow’s intersections. *IEEE Transactions on Vehicular Technology*, 71(7):7033–7043, 2022.
- Peter Auer, Nicolo Cesa-Bianchi, Yoav Freund, and Robert E Schapire. The nonstochastic multiarmed bandit problem. *SIAM journal on computing*, 32(1):48–77, 2002.
- Didier Aussel and Anton Svensson. A short state of the art on multi-leader-follower games. *Bilevel optimization: Advances and next challenges*, pages 53–76, 2020.
- Yu Bai, Chi Jin, and Tiancheng Yu. Near-optimal reinforcement learning with self-play. *Advances in neural information processing systems*, 33:2159–2170, 2020.
- Yu Bai, Chi Jin, Huan Wang, and Caiming Xiong. Sample-efficient learning of stackelberg equilibria in general-sum games. *Advances in Neural Information Processing Systems*, 34: 25799–25811, 2021.
- James P Bailey and Georgios Piliouras. Multiplicative weights update in zero-sum games. In *Proceedings of the 2018 ACM Conference on Economics and Computation*, pages 321–338, 2018.
- Maria-Florina Balcan, Avrim Blum, Nika Haghtalab, and Ariel D Procaccia. Commitment without regrets: Online learning in stackelberg security games. In *Proceedings of the sixteenth ACM conference on economics and computation*, pages 61–78, 2015.
- Avrim Blum, Nika Haghtalab, and Ariel D Procaccia. Learning optimal commitment to overcome insecurity. *Advances in Neural Information Processing Systems*, 27, 2014.
- Michele Breton, Abderrahmane Alj, and Alain Haurie. Sequential stackelberg equilibria in two-person games. *Journal of Optimization Theory and Applications*, 59(1):71–97, 1988.
- Yang Cai, Haipeng Luo, Chen-Yu Wei, and Weiqiang Zheng. Uncoupled and convergent learning in two-player zero-sum markov games with bandit feedback. *Advances in Neural Information Processing Systems*, 36:36364–36406, 2023.
- Yang Cai, Haipeng Luo, Chen-Yu Wei, and Weiqiang Zheng. Near-optimal policy optimization for correlated equilibrium in general-sum markov games. In *International Conference on Artificial Intelligence and Statistics*, pages 3889–3897. PMLR, 2024.

- Yang Cai, Haipeng Luo, Chen-Yu Wei, and Weiqiang Zheng. From average-iterate to last-iterate convergence in games: A reduction and its applications. *arXiv preprint arXiv:2506.03464*, 2025.
- Siyu Chen, Mengdi Wang, and Zhuoran Yang. Actions speak what you want: Provably sample-efficient reinforcement learning of the quantal stackelberg equilibrium from strategic feedbacks. *arXiv preprint arXiv:2307.14085*, 2023.
- Vincent Conitzer and Tuomas Sandholm. Computing the optimal strategy to commit to. In *Proceedings of the 7th ACM conference on Electronic commerce*, pages 82–90, 2006.
- Christos Dimitrakakis, David C Parkes, Goran Radanovic, and Paul Tylkin. Multi-view decision processes: the helper-ai problem. *Advances in neural information processing systems*, 30, 2017.
- Jinshuo Dong, Aaron Roth, Zachary Schutzman, Bo Waggoner, and Zhiwei Steven Wu. Strategic classification from revealed preferences. In *Proceedings of the 2018 ACM Conference on Economics and Computation*, pages 55–70, 2018.
- Liad Erez, Tal Lancewicki, Uri Sherman, Tomer Koren, and Yishay Mansour. Regret minimization and convergence to equilibria in general-sum markov games. In *International Conference on Machine Learning*, pages 9343–9373. PMLR, 2023.
- Jerzy Filar and Koos Vrieze. *Competitive Markov decision processes*. Springer Science & Business Media, 2012.
- Dylan J Foster, Noah Golowich, and Sham M Kakade. Hardness of independent learning and sparse equilibrium computation in markov games. In *International Conference on Machine Learning*, pages 10188–10221. PMLR, 2023.
- Yoav Freund and Robert E Schapire. Adaptive game playing using multiplicative weights. *Games and Economic Behavior*, 29(1-2):79–103, 1999.
- Matthias Gerstgrasser and David C Parkes. Oracles & followers: Stackelberg equilibria in deep multi-agent reinforcement learning. In *International Conference on Machine Learning*, pages 11213–11236. PMLR, 2023.
- Arman Ghasemi, Amin Shojaeighadikolaei, Kailani Jones, Morteza Hashemi, Alexandru G Bardas, and Reza Ahmadi. A multi-agent deep reinforcement learning approach for a distributed energy marketplace in smart grids. In *2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, pages 1–6. IEEE, 2020.
- Denizalp Goktas, Sadie Zhao, and Amy Greenwald. Zero-sum stochastic stackelberg games. *Advances in Neural Information Processing Systems*, 35:11658–11672, 2022.
- Albert Xin Jiang, Ariel D Procaccia, Yundi Qian, Nisarg Shah, and Milind Tambe. Defender (mis) coordination in security games. *Proceedings of the Twenty-Third International Joint Conference on Artificial Intelligence*, 2013.
- Chi Jin, Tiancheng Jin, Haipeng Luo, Suvrit Sra, and Tiancheng Yu. Learning adversarial markov decision processes with bandit feedback and unknown transition. In *International Conference on Machine Learning*, pages 4860–4869. PMLR, 2020.
- Chi Jin, Qinghua Liu, Yuanhao Wang, and Tiancheng Yu. V-learning—a simple, efficient, decentralized algorithm for multiagent rl. *arXiv preprint arXiv:2110.14555*, 2021.
- Junqi Jin, Chengru Song, Han Li, Kun Gai, Jun Wang, and Weinan Zhang. Real-time bidding with multi-agent reinforcement learning in display advertising. In *Proceedings of the 27th ACM international conference on information and knowledge management*, pages

- 2193–2201, 2018.
- Tiancheng Jin, Junyan Liu, Chloé Rouyer, William Chang, Chen-Yu Wei, and Haipeng Luo. No-regret online reinforcement learning with adversarial losses and transitions. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 38520–38585. Curran Associates, Inc., 2023.
- Sham Kakade and John Langford. Approximately optimal approximate reinforcement learning. In *Proceedings of the nineteenth international conference on machine learning*, pages 267–274, 2002.
- Katarzyna Kańska and Agnieszka Wiszniewska-Matyszek. Dynamic stackelberg duopoly with sticky prices and a myopic follower. *Operational Research*, 22(4):4221–4252, 2022.
- Hsu Kao, Chen-Yu Wei, and Vijay Subramanian. Decentralized cooperative reinforcement learning with hierarchical information structure. In *International Conference on Algorithmic Learning Theory*, pages 573–605. PMLR, 2022.
- Robert Kleinberg and Tom Leighton. The value of knowing a demand curve: Bounds on regret for online posted-price auctions. In *44th Annual IEEE Symposium on Foundations of Computer Science, 2003. Proceedings.*, pages 594–605. IEEE, 2003.
- Tor Lattimore and Csaba Szepesvári. *Bandit algorithms*. Cambridge University Press, 2020.
- Niklas Lauffer, Mahsa Ghasemi, Abolfazl Hashemi, Yagiz Savas, and Ufuk Topcu. No-regret learning in dynamic stackelberg games. *IEEE Transactions on Automatic Control*, 69(3): 1418–1431, 2023.
- Stefanos Leonardos, Will Overman, Ioannis Panageas, and Georgios Piliouras. Global convergence of multi-agent policy gradient in markov potential games. *arXiv preprint arXiv:2106.01969*, 2021.
- Joshua Letchford, Vincent Conitzer, and Kamesh Munagala. Learning and approximating the optimal strategy to commit to. In *International symposium on algorithmic game theory*, pages 250–262. Springer, 2009.
- Yuanguo Lin, Yong Liu, Fan Lin, Lixin Zou, Pengcheng Wu, Wenhua Zeng, Huanhuan Chen, and Chunyan Miao. A survey on reinforcement learning for recommender systems. *IEEE Transactions on Neural Networks and Learning Systems*, 35(10):13164–13184, 2023.
- Víctor Bucarey López, Eugenio Della Vecchia, Alain Jean-Marie, and Fernando Ordoñez. Stationary strong stackelberg equilibrium in discounted stochastic games. *IEEE Transactions on Automatic Control*, 68(9):5271–5286, 2022.
- Haipeng Luo, Chen-Yu Wei, and Chung-Wei Lee. Policy optimization in adversarial mdps: Improved exploration via dilated bonuses. *Advances in Neural Information Processing Systems*, 34:22931–22942, 2021.
- Odalric-Ambrym Maillard and Shie Mannor. Latent bandits. In *International Conference on Machine Learning*, pages 136–144. PMLR, 2014.
- Weichao Mao and Tamer Başar. Provably efficient reinforcement learning in decentralized general-sum markov games. *Dynamic Games and Applications*, 13(1):165–186, 2023.
- Jason R Marden and Jeff S Shamma. Game theory and distributed control. In *Handbook of game theory with economic applications*, volume 4, pages 861–899. Elsevier, 2015.
- Janusz Marecki, Gerry Tesauro, and Richard Segal. Playing repeated stackelberg games with unknown opponents. In *Proceedings of the 11th International Conference on Autonomous Agents and Multiagent Systems-Volume 2*, pages 821–828, 2012.

- Panayotis Mertikopoulos, Christos Papadimitriou, and Georgios Piliouras. Cycles in adversarial regularized learning. In *Proceedings of the twenty-ninth annual ACM-SIAM symposium on discrete algorithms*, pages 2703–2717. SIAM, 2018.
- Stefanos Nikolaidis, Swaprava Nath, Ariel D Procaccia, and Siddhartha Srinivasa. Game-theoretic modeling of human adaptation in human-robot collaboration. In *Proceedings of the 2017 ACM/IEEE international conference on human-robot interaction*, pages 323–331, 2017.
- Francesco Orabona. A modern introduction to online learning. *arXiv preprint arXiv:1912.13213*, 2019.
- Arvind Patel and Anamika Mishra. Intelligent bargaining agents in digital marketplaces: A fusion of reinforcement learning and game-theoretic principles. *International Journal of Advanced Artificial Intelligence Research*, 2(03):6–12, 2025.
- Yagna Patel. Optimizing market making using multi-agent reinforcement learning. *arXiv preprint arXiv:1812.10252*, 2018.
- Binghui Peng, Weiran Shen, Pingzhong Tang, and Song Zuo. Learning optimal strategies to commit to. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, 2019.
- Goran Radanovic, Rati Devidze, David Parkes, and Adish Singla. Learning to collaborate in markov decision processes. In *International Conference on Machine Learning*, pages 5261–5270. PMLR, 2019.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Shai Shalev-Shwartz, Shaked Shammah, and Amnon Shashua. Safe, multi-agent, reinforcement learning for autonomous driving. *arXiv preprint arXiv:1610.03295*, 2016.
- Jeff S Shamma and Gürdal Arslan. Dynamic fictitious play, dynamic gradient play, and distributed convergence to nash equilibria. *IEEE Transactions on Automatic Control*, 50(3):312–327, 2005.
- Lloyd S Shapley. Stochastic games. *Proceedings of the national academy of sciences*, 39(10):1095–1100, 1953.
- Matt Shum. Lecture 7: Incumbent advantage. entry deterrence and accommodation. Presentation, EC 105: Industrial Organization, California Institute of Technology, 2023.
- Arunesh Sinha, Fei Fang, Bo An, Christopher Kiekintveld, and Milind Tambe. Stackelberg security games: Looking beyond a decade of success. *IJCAI*, 2018.
- Ziang Song, Song Mei, and Yu Bai. When can we learn general-sum markov games with a large number of players sample-efficiently? *arXiv preprint arXiv:2110.04184*, 2021.
- Heinrich von Stackelberg. Marktform und gleichgewicht. (*No Title*), 1934.
- Jingrui Sun, Hanxiao Wang, and Jiaqiang Wen. Zero-sum stackelberg stochastic linear-quadratic differential games. *SIAM Journal on Control and Optimization*, 61(1):252–284, 2023.
- Milind Tambe. *Security and game theory: algorithms, deployed systems, lessons learned*. Cambridge university press, 2011.
- Yi Tian, Yuanhao Wang, Tiancheng Yu, and Suvrit Sra. Online learning in unknown markov games. In *International conference on machine learning*, pages 10279–10288. PMLR, 2021.
- Stelios Triantafyllou, Aleksa Sukovic, Yasaman Zolfimoselo, and Goran Radanovic. Counterfactual effect decomposition in multi-agent sequential decision making. *arXiv preprint*

- arXiv:2410.12539*, 2024.
- John von Neumann. *Mathematische annalen* 100 (1928), pp. *Contributions to the Theory of Games*, 100(40):13, 1959.
- Yuanhao Wang, Qinghua Liu, Yu Bai, and Chi Jin. Breaking the curse of multiagency: Provably efficient decentralized multi-agent rl with function approximation. In *The Thirty Sixth Annual Conference on Learning Theory*, pages 2793–2848. PMLR, 2023.
- Chen-Yu Wei, Chung-Wei Lee, Mengxiao Zhang, and Haipeng Luo. Last-iterate convergence of decentralized optimistic gradient descent/ascent in infinite-horizon competitive markov games. In *Conference on learning theory*, pages 4259–4299. PMLR, 2021.
- Yuepeng Yang and Cong Ma. $o(t^{-1})$ convergence of optimistic-follow-the-regularized-leader in two-player zero-sum markov games. *arXiv preprint arXiv:2209.12430*, 2022.
- Yaolong Yu and Haipeng Chen. Decentralized online learning in general-sum stackelberg games. *arXiv preprint arXiv:2405.03158*, 2024.
- Yaolong Yu, Haifeng Xu, and Haipeng Chen. Learning correlated stackelberg equilibrium in general-sum multi-leader-single-follower games. *arXiv preprint arXiv:2210.12470*, 2022.
- Runyu Zhang, Qinghua Liu, Huan Wang, Caiming Xiong, Na Li, and Yu Bai. Policy optimization for markov games: Unified framework and faster convergence. *Advances in Neural Information Processing Systems*, 35:21886–21899, 2022.
- Geng Zhao, Banghua Zhu, Jiantao Jiao, and Michael Jordan. Online learning in stackelberg games with an omniscient follower. In *International Conference on Machine Learning*, pages 42304–42316. PMLR, 2023.
- Han Zhong, Zhuoran Yang, Zhaoran Wang, and Michael I Jordan. Can reinforcement learning find stackelberg-nash equilibria in general-sum markov games with myopically rational followers? *Journal of Machine Learning Research*, 24(35):1–52, 2023.
- Banghua Zhu, Stephen Bates, Zhuoran Yang, Yixin Wang, Jiantao Jiao, and Michael I Jordan. The sample complexity of online contract design. *arXiv preprint arXiv:2211.05732*, 2022.

A Implementation Details

A.1 Algorithm 1: Policy Optimization with α Exploration and Upper Confidence Bound

Here we include the details on computing the transition confidence set and occupancy bounds for the leader in Algorithm 1. We have included the efficient COMP-UOB (Algorithm 2) and COMP-LOB (Algorithm 3) procedures from Luo et al. [2021] and Jin et al. [2020] to compute the state-action occupancy bounds $\bar{q}$ and $\underline{q}$, both which use the GREEDY algorithm (Algorithm 4). Note that the unmodified π is drawn from the exponential weights algorithm, but we use $\tilde{\pi}$ to compute the occupancy bounds.

Transition confidence set. Let k be the new epoch number. Compute the new empirical transition $\bar{P}_k(s'|s, a) := \frac{N_k(s, a, s')}{\max\{1, N_k(s, a)\}}$ and confidence set $\mathcal{P}_k$:

$$\mathcal{P}_k = \left\{ \hat{P} : \left| \hat{P}(s' | s, a) - \bar{P}_k(s' | s, a) \right| \leq \text{conf}_k(s' | s, a), \quad \forall (s, a, s') \in \mathcal{S}_h \times \mathcal{A} \times \mathcal{S}_{h+1}, \quad h = 0, 1, \dots, H-1 \right\}. \quad (9)$$

$$\text{where } \text{conf}_k(s' | s, a) = 4 \sqrt{\frac{\bar{P}_k(s' | s, a) \log\left(\frac{TSA}{\delta}\right)}{\max\{1, N_k(s, a)\}}} + \frac{28 \log\left(\frac{TSA}{\delta}\right)}{3 \max\{1, N_k(s, a)\}}.$$

Algorithm 2 COMP-UOB (Algorithm 3 of Jin et al. [2020], Algorithm 4 of Luo et al. [2021])

Require: A policy $\tilde{\pi}_t$, a state-action pair (s, a) , and a confidence set $\mathcal{P}$ of the form

$$\left\{ \hat{P} : \left| \hat{P}(s' | s, a) - \bar{P}(s' | s, a) \right| \leq \epsilon(s' | s, a), \quad \forall (s, a, s') \right\}$$

- 1: **Initialize:** for all $\tilde{s} \in \mathcal{S}_{h(s)}$, set $f(\tilde{s}) = \mathbb{1}\{\tilde{s} = s\}$
- 2: **for** $h = h(s) - 1$ to 0 **do**
- 3: **for all** $\tilde{s} \in \mathcal{S}_h$ **do**
- 4: Compute $f(\tilde{s})$ based on:

$$f(\tilde{s}) = \sum_{a \in \mathcal{A}} \tilde{\pi}_t(a | \tilde{s}) \cdot \text{GREEDY}(f, \bar{P}(\cdot | \tilde{s}, a), \epsilon(\cdot | \tilde{s}, a), \max)$$

- 5: **end for**
 - 6: **end for**
 - 7: **return** $\tilde{\pi}_t(a | s) f(s_0)$
-

Algorithm 3 COMP-LOB (Algorithm 5 of Luo et al. [2021])

Require: A policy $\tilde{\pi}_t$, a state-action pair (s, a) , and a confidence set $\mathcal{P}$ of the form

$$\left\{ \hat{P} : \left| \hat{P}(s'|s, a) - \bar{P}(s'|s, a) \right| \leq \epsilon(s'|s, a), \forall (s, a, s') \right\}$$

- 1: **Initialize:** for all $\tilde{s} \in S_{h(s)}$, set $f(\tilde{s}) = \mathbb{1}\{\tilde{s} = s\}$
- 2: **for** $k = k(s) - 1$ **to** 0 **do**
- 3: **for all** $\tilde{s} \in S_h$ **do**
- 4: Compute $f(\tilde{s})$ based on:

$$f(\tilde{s}) = \sum_{a \in A} \tilde{\pi}_t(a|\tilde{s}) \cdot \text{GREEDY}(f, \bar{P}(\cdot|\tilde{s}, a), \epsilon(\cdot|\tilde{s}, a), \min)$$

- 5: **end for**
 - 6: **end for**
 - 7: **return** $\tilde{\pi}_t(a|s)f(s_0)$
-

Algorithm 4 GREEDY (Algorithm 6 of Luo et al. [2021])

Require: $f : S \rightarrow [0, 1]$, a distribution $\bar{p}$ over n states of layer h , positive numbers $\{\epsilon(s)\}_{s \in S_h}$, objective OPTIMIZE (MAX for COMP-UOB and MIN for COMP-LOB)

- 1: **Initialize:** $j^- = 1$, $j^+ = n$, sort $\{f(s)\}_{s \in S_h}$ and find σ such that
 - 2: **if** OPTIMIZE = max **then**
 - 3: $f(\sigma(1)) \leq f(\sigma(2)) \leq \dots \leq f(\sigma(n))$
 - 4: **else**
 - 5: $f(\sigma(1)) \geq f(\sigma(2)) \geq \dots \geq f(\sigma(n))$
 - 6: **end if**
 - 7: **while** $j^- < j^+$ **do**
 - 8: $s^- = \sigma(j^-)$, $s^+ = \sigma(j^+)$
 - 9: $\delta = \min\{\bar{p}(s^-), \epsilon(s^-)\}$
 - 10: $\delta^+ = \min\{1 - \bar{p}(s^+), \epsilon(s^+)\}$
 - 11: $\bar{p}(s^-) \leftarrow \bar{p}(s^-) - \min\{\delta, \delta^+\}$
 - 12: $\bar{p}(s^+) \leftarrow \bar{p}(s^+) + \min\{\delta, \delta^+\}$
 - 13: **if** $\delta < \delta^+$ **then**
 - 14: $\epsilon(s^+) \leftarrow \epsilon(s^+) - \delta$
 - 15: $j^- \leftarrow j^- + 1$
 - 16: **else**
 - 17: $\epsilon(s^-) \leftarrow \epsilon(s^-) - \delta^+$
 - 18: $j^+ \leftarrow j^+ - 1$
 - 19: **end if**
 - 20: **end while**
 - 21: **return** $\sum_{j=1}^n \bar{p}(\sigma(j))f(\sigma(j))$
-

B Auxiliary Results

Lemma 1 (Corollary 5.5 from Lattimore and Szepesvári [2020]) *Assume that $X_i - \mu$ are independent, 1-subgaussian random variables. Then for any $\varepsilon > 0$,*

$$\mathbb{P}(|\hat{\mu} - \mu| > \varepsilon) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2}\right)$$

where $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$.

Lemma 2 (Azuma-Hoeffding for MDS) *Let $(\xi_t)_{t \in [T]}$ be a martingale difference sequence adapted to a filtration $(\mathcal{F}_t)_{t \in [T]}$ satisfying*

$$|\xi_t| \leq H_t \quad \text{a.s. for all } t \in [T].$$

Then, for any $\varepsilon > 0$, it holds:

$$\mathbb{P}\left(\left|\sum_{t=1}^T \xi_t\right| \geq \varepsilon\right) \leq 2 \exp\left(-\frac{\varepsilon^2}{2 \sum_{t=1}^T H_t^2}\right)$$

Lemma 3 (Exponential Weights Algorithm) *Let $\eta > 0$, and let $\pi_t \in \Delta(\mathcal{A})$ and $\ell_t \in \mathbb{R}^A$ satisfy, for all $t \in [T]$ and $a \in \mathcal{A}$,*

$$\begin{aligned} \pi_1(a) &= \frac{1}{A}, \\ \pi_{t+1}(a) &= \frac{\pi_t(a) \exp(-\eta \ell_t(a))}{\sum_{a' \in \mathcal{A}} \pi_t(a') \exp(-\eta \ell_t(a'))}, \\ |\eta \ell_t(a)| &\leq 1. \end{aligned}$$

Then for any $\pi^ \in \Delta(\mathcal{A})$,*

$$\sum_{t=1}^T \sum_{a \in \mathcal{A}} (\pi_t(a) - \pi^*(a)) \ell_t(a) \leq \frac{\ln |\mathcal{A}|}{\eta} + \eta \sum_{t=1}^T \sum_{a \in \mathcal{A}} \pi_t(a) \ell_t(a)^2. \quad (10)$$

Lemma 4 (Lemma 3.1 from Luo et al. [2021]) *Let $b_t(s, a)$ be a non-negative loss function, and let $\hat{P}$ be a transition function. Suppose that the following holds for all s, a :*

$$\begin{aligned} B_t(s, a) &= b_t(s, a) + \left(1 + \frac{1}{H}\right) \mathbb{E}_{s' \sim \hat{P}(\cdot|s, a)} \mathbb{E}_{a' \sim \pi_t(\cdot|s')} [B_t(s', a')] \\ &\geq b_t(s, a) + \left(1 + \frac{1}{H}\right) \mathbb{E}_{s' \sim P(\cdot|s, a)} \mathbb{E}_{a' \sim \pi_t(\cdot|s')} [B_t(s', a')]. \end{aligned} \quad (19)$$

with $B_t(s_H, a) \triangleq 0$, and suppose that

$$\begin{aligned} & \sum_s q^*(s) \sum_{t=1}^T \sum_a (\pi_t(a|s) - \pi^*(a|s)) (Q_t^{\tilde{\pi}_t}(s, a) - B_t(s, a)) \\ & \leq o(T) + \sum_{t=1}^T V^{\pi^*}(s_0; b_t) + \frac{1}{H} \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a|s) B_t(s, a). \end{aligned}$$

holds. Then

$$\sum_s q^*(s) \sum_{t=1}^T \sum_a (\pi_t(a|s) - \pi^*(a|s)) Q_t^{\tilde{\pi}_t}(s, a) \leq o(T) + 3 \sum_{t=1}^T \hat{V}^{\pi_t}(s_0; b_t). \quad (11)$$

Proof: Note that the expectation over actions in the next state for the dilated bonus calculation are taken with respect to the unmodified policy π , *not* $\tilde{\pi}$. Applying the same recursive techniques in Lemma B.1 of Luo et al. [2021] thus achieves the conclusion. $\square$

C Proof of Theorem 1

As detailed in Section 4, we first decompose the Stackelberg MDP regret of the leader into the following three terms:

$$\begin{aligned} R_{\text{Stack}}(T) &= \sum_{t=1}^T \left(V^{\pi_t, \pi_*^f}(s_0) - V^{\pi_*, \pi_*^f}(s_0) \right) \\ &= \sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_*^f}(s_0) - V^{\tilde{\pi}_t, \pi_t^f}(s_0) \right) && \text{(Term I)} \\ &+ \sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_t^f}(s_0) - V^{\pi_*, \pi_t^f}(s_0) \right) && \text{(Term II)} \\ &+ \sum_{t=1}^T \left(V^{\pi_*, \pi_t^f}(s_0) - V^{\pi_*, \pi_*^f}(s_0) \right) && \text{(Term III)} \end{aligned}$$

Here, Term I intuitively captures the total difference in the *leader's losses* for episodes in which the follower played some policy π_t^f versus the best response policy π_*^f . Term II captures the leader's total incurred regret from playing some policy π_t other than the optimal policy π_* with the follower playing $\{\pi_t^f\}$. Finally, Term III captures the difference in the leader's losses of the *optimal leader policy* π_* between rounds in which the follower played some policy π_t^f versus the best response policy π_*^f . From a high-level perspective, Terms I and III capture the learning capacity of the follower, using results from UCB error analysis, and Term II

captures the learning of the leader, using regret guarantees from Luo et al. [2021]. Term III also critically uses the α exploration, ensuring sufficient learning for the follower for any particular state-leader action environment. We first develop some auxiliary lemmas that will aid in bounding the terms above.

C.1 Bounding $P((s_h, a_h) = (s, a))$ for given step h and (s, a) (Exploration Guarantee)

Lemma 5 *Let $\beta := \min\{P(s_{h+1}|s_h, a) : (s_h, a, s_{h+1}) \in \mathcal{S}_h \times \mathcal{A} \times \mathcal{S}_{h+1}, h \in \{0, 1, \dots, H-1\}\}$ be the minimum transition probability gap (nonzero by Assumption 3). Then, using Algorithm 1 yields the lower bound for the probability of visiting any $(s, a) \in \mathcal{S} \times \mathcal{A}$ in any given layer h as:*

$$\mathbb{P}((s_h, a_h) = (s, a)) \geq \frac{\alpha\beta}{A}$$

Proof: We write:

$$\begin{aligned} \mathbb{P}((s_h, a_h) = (s, a)) &= \pi_t(a|s) \cdot \mathbb{P}(s_h = s) \geq \left(\frac{\alpha}{A}\right) \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} P(s|(s', a')) \cdot \mathbb{P}((s_{h-1}, a_{h-1}) = (s', a')) \\ &\geq \left(\frac{\alpha}{A}\right) \beta \cdot \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} \mathbb{P}((s_{h-1}, a_{h-1}) = (s', a')) \\ &= \frac{\alpha\beta}{A} \end{aligned}$$

□

C.2 Bounding $n_k^f(n_{(s,a)}(T))$ for suboptimal responses $k \neq \pi_\star^f(s, a)$

Recall that $n_k^f(n_{(s,a)}(t))$ is the number of pulls of follower action $k \in \mathcal{K}$ as a function of the number of times state-leader action pair (s, a) is visited up to episode t . We extend upon results from Orabona [2019] and Yu et al. [2022] and using the protocol of the follower's UCB algorithm, we have the following lemma:

Lemma 6 *Let ε be the minimum suboptimality gap, that is, $\varepsilon = \min_{s \in \mathcal{S}, a \in \mathcal{A}, k \neq \pi_\star^f(s, a)} \Delta_{s,a,k}$, where $\Delta_{s,a,k} = l^f(s, a, k) - l^f(s, a, \pi_\star^f(s, a))$. Using Algorithm 1, we have with probability at least $1 - \delta$ that the number of pulls of any suboptimal follower response $k \neq \pi_\star^f(s, a)$ to any state-leader action (s, a) is bounded as (suppressing constants):*

$$n_k^f(n_{(s,a)}(T)) \leq \mathcal{O}\left(\frac{\log T}{\varepsilon^2} + \log\left(\frac{SAK}{\delta}\right)\right)$$

Proof: From Orabona [2019], we have that the expected number of pulls of any suboptimal follower response $k \neq \pi_\star^f(s, a)$ to some state-leader action (s, a) is bounded as:

$$\mathbb{E} \left[n_k^f(n_{(s,a)}(T)) \right] \leq \mathcal{O} \left(\frac{24 \log n_{(s,a)}(T)}{\varepsilon^2} \right) \quad (12)$$

Furthermore, suppose state-leader action (s, a) at some step h is visited at rounds $t_1, t_2, \dots, t_{n_{(s,a)}(T)} \in [T]$, and let $k \neq \pi_\star^f(s, a)$. If $n_k^f(n_{(s,a)}(T-1)) \leq \frac{24 \log n_{(s,a)}(T)}{\Delta_{s,a,k}^2}$, then $n_k^f(n_{(s,a)}(T)) \leq \mathcal{O} \left(\frac{24 \log n_{(s,a)}(T)}{\varepsilon^2} \right)$, and the claim holds. Thus, suppose that $n_k^f(n_{(s,a)}(T-1)) > \frac{24 \log n_{(s,a)}(T)}{\Delta_{s,a,k}^2}$, and so let $j \in [n_{(s,a)}(T)]$ be the largest index such that $n_k^f(n_{(s,a)}(t_{j-1})) \leq \frac{24 \log n_{(s,a)}(T)}{\Delta_{s,a,k}^2}$. By Orabona [2019], one has that at any round $t_i > t_j$, $i \in [n_{(s,a)}(T)]$:

$$\mathbb{P} \left(s_{t_i,h} = s, a_{t_i,h} = a, \pi_{t_i}^f(s, a) = k \right) \leq 2(i-1)i^{-3}.$$

Following the line of Lemma E.1 from Yu et al. [2022], we see that with probability at least $1 - \frac{\delta}{SAK}$,

$$\sum_{i=j+1}^{n_{(s,a)}(T)} \mathbb{I} \left(s_{t_i,h} = s, a_{t_i,h} = a, \pi_{t_i}^f(s, a) = k \right) \leq \mathcal{O} \left(\log \left(\frac{1}{\delta} \right) + \log(SAK) \right) \quad (13)$$

And finally, using the union bound over all $(s, a) \in \mathcal{S} \times \mathcal{A}$ and $k \neq \pi_\star^f(s, a)$, we get with probability at least $1 - \delta$ that for any (s, a, k) :

$$n_k^f(n_{(s,a)}(T)) \leq n_k^f(n_{(s,a)}(t_{j-1})) + 1 + \sum_{i=j+1}^{n_{(s,a)}(T)} \mathbb{I} \left(s_{t_i,h} = s, a_{t_i,h} = a, \pi_{t_i}^f(s, a) = k \right) \quad (14)$$

$$\leq \frac{24 \log n_{(s,a)}(T)}{\Delta_{s,a,k}^2} + 1 + \mathcal{O} \left(\log \left(\frac{SAK}{\delta} \right) \right) \quad (15)$$

$$\leq \mathcal{O} \left(\frac{\log T}{\varepsilon^2} + \log \left(\frac{SAK}{\delta} \right) \right) \quad (16)$$

□

Now we proceed to analyze the regret terms.

C.3 Bound on Term I

Lemma 7 *With the players following Algorithm 1 with $\alpha = \mathcal{O}(H^{-1}S^{\frac{1}{2}}A(\log A)^{\frac{1}{3}}T^{-\frac{1}{4}})$, we have that with probability at least $1 - 2\delta$ (suppressing constants),*

$$\underbrace{\sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_*^f}(s_0) - V^{\tilde{\pi}_t, \pi_t^f}(s_0) \right)}_{\text{Term I}} \leq \mathcal{O} \left(SAK \left(\frac{\log T}{\varepsilon^2} + \log \left(\frac{SAK}{\delta} \right) \right) + H \sqrt{T \log \left(\frac{1}{\delta} \right)} \right)$$

Proof: For each episode, define the mean-loss difference

$$d_t(s, a) := l(s, a, \pi_f^*(s, a)) - l(s, a, \pi_t^f(s, a)).$$

Conditional on the pre-episode history $\mathcal{F}_{t-1}$, both $\tilde{\pi}_t$ and π_t^f are fixed. Assumption 1 implies that the follower action does not affect the state-leader-action trajectory distribution. Hence

$$\mathbb{E} \left[\sum_{h=0}^{H-1} d_t(s_{t,h}, a_{t,h}) \middle| \mathcal{F}_{t-1} \right] = V^{\tilde{\pi}_t, \pi_f^*}(s_0) - V^{\tilde{\pi}_t, \pi_t^f}(s_0).$$

Therefore

$$\xi_t := V^{\tilde{\pi}_t, \pi_f^*}(s_0) - V^{\tilde{\pi}_t, \pi_t^f}(s_0) - \sum_{h=0}^{H-1} d_t(s_{t,h}, a_{t,h})$$

is a martingale difference sequence with $|\xi_t| \leq 2H$. Moreover, because $l \in [0, 1]$,

$$d_t(s, a) \leq \mathbb{I}\{\pi_t^f(s, a) \neq \pi_f^*(s, a)\}.$$

Lemma 6 and a union bound over (s, a, k) therefore imply, with probability at least $1 - \delta$,

$$\sum_{t=1}^T \sum_{h=0}^{H-1} d_t(s_{t,h}, a_{t,h}) \leq \mathcal{O} \left(SAK \left(\frac{\log T}{\varepsilon^2} + \log \frac{SAK}{\delta} \right) \right).$$

Azuma–Hoeffding applied to (ξ_t) contributes $\mathcal{O}(H\sqrt{T \log(1/\delta)})$. Combining the two bounds and union bounding the corresponding good events proves the lemma. $\square$

C.4 Bound on Term II

Lemma 8 *With the players following Algorithm 1 with $\alpha = \mathcal{O}(H^{-1}S^{\frac{1}{2}}A(\log A)^{\frac{1}{3}}T^{-\frac{1}{4}})$, we have that with probability at least $1 - \mathcal{O}(\delta)$,*

$$\underbrace{\sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_t^f}(s_0) - V^{\pi_*, \pi_t^f}(s_0) \right)}_{\text{Term II}} \leq \tilde{\mathcal{O}} \left(H^4 + H^3 S A^{\frac{1}{2}} T^{\frac{3}{4}} + H S^{\frac{1}{2}} A T^{\frac{3}{4}} \right)$$

where $\tilde{\mathcal{O}}$ suppresses logarithmic factors.

Proof: Here, apply regret techniques given in Luo et al. [2021] for their policy optimization algorithm while accounting for the additional incurred regret from the explicit exploration parameter α . As a reminder, $\tilde{\pi} = (1 - \alpha)\pi + \alpha[1/A, \dots, 1/A]$ is a linear combination of π and a uniform probability across $\mathcal{A}$.

We first formulate an adversarial environment for the leader by redefining $l(s, a, \pi_t^f(s, a)) = l_t(s, a)$ (and the noisy observed losses, $\ell(s, a, \pi_t^f(s, a)) = \ell_t(s, a)$) for all $(s, a) \in \mathcal{S} \times \mathcal{A}$. Thus, we "eliminate" the dependence of the leader's losses on the follower's policy by treating the leader's losses as adversarial. We can redefine adversarial value functions and Bellman equations for the leader as such:

$$\begin{cases} V_t^\pi(s_{h'}) = V^\pi(s_{h'}; \ell_t) := \mathbb{E}_{a_h \sim \pi(\cdot|s_h), s_{h+1} \sim P(\cdot|s_h, a_h)} \left[\sum_{h=h'}^{H-1} \ell_t(s_h, a_h) \right] \\ Q_t^\pi(s_h, a_h) = Q^\pi(s_h, a_h; \ell_t) := \ell_t(s_h, a_h) + \mathbb{E}_{s_{h+1} \sim P(\cdot|s_h, a_h)} [V_t^\pi(s_{h+1})] \\ V_t^\pi(s_h) = \mathbb{E}_{a_h \sim \pi(\cdot|s_h)} [Q_t^\pi(s_h, a_h)] \end{cases} \quad (17)$$

Using the adversarial value functions, we can rewrite and bound Term II as:

$$\underbrace{\sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_t^f}(s_0) - V^{\pi_*, \pi_t^f}(s_0) \right)}_{\text{Term II}} = \sum_{t=1}^T V_t^{\tilde{\pi}_t}(s_0) - \sum_{t=1}^T V_t^{\pi_*}(s_0) \leq \sum_{t=1}^T V_t^{\tilde{\pi}_t}(s_0) - \min_{\pi} \sum_{t=1}^T V_t^{\pi}(s_0) = R(T) \quad (18)$$

where $R(T)$ is the (non-Stackelberg) regret of the traditional adversarial MDP setting. For notational purposes, we change $\pi_* := \arg \min_{\pi} \sum_{t=1}^T V_t^{\pi}(s_0)$ for the rest of this lemma's proof only. Furthermore, as the Stackelberg Markov game is leader-controlled, we let $q^\pi(s)$ be the probability of visiting state s following policy π . Now, applying the performance-difference lemma to $R(T)$ (Luo et al. [2021], Kakade and Langford [2002]) with leader policies $\tilde{\pi}_t$ (given

by Algorithm 1) and π_* , we have:

$$R(T) = \sum_{s \in \mathcal{S}} q^{\pi_*}(s) \sum_{t=1}^T \sum_{a \in \mathcal{A}} (\tilde{\pi}_t(a|s) - \pi_*(a|s)) Q_t^{\tilde{\pi}_t}(s, a). \quad (19)$$

Then, recalling the definition of $\tilde{\pi}$ gives:

$$\begin{aligned} R(T) &= \sum_{s \in \mathcal{S}} q^{\pi_*}(s) \sum_{t=1}^T \sum_{a \in \mathcal{A}} \left((1 - \alpha) \pi_t(a|s) + \frac{\alpha}{A} - \pi_*(a|s) \right) Q_t^{\tilde{\pi}_t}(s, a) \\ &= (1 - \alpha) \sum_{s \in \mathcal{S}} q^{\pi_*}(s) \sum_{t=1}^T \sum_{a \in \mathcal{A}} (\pi_t(a|s) - \pi^*(a|s)) Q_t^{\tilde{\pi}_t}(s, a) \\ &\quad + \alpha \sum_{s \in \mathcal{S}} q^{\pi_*}(s) \sum_{t=1}^T \sum_{a \in \mathcal{A}} \left(\frac{1}{A} - \pi^*(a|s) \right) Q_t^{\tilde{\pi}_t}(s, a) \\ &\leq \underbrace{\sum_{s \in \mathcal{S}} q^{\pi_*}(s) \sum_{t=1}^T \sum_{a \in \mathcal{A}} (\pi_t(a|s) - \pi^*(a|s)) Q_t^{\tilde{\pi}_t}(s, a)}_{(\star)} + \mathcal{O}(\alpha T H^2), \end{aligned}$$

where in the last inequality, we used a trivial bound of $Q_t^{\tilde{\pi}} \leq H$ and $\sum_s q^{\pi_*}(s) = H$ to get $\mathcal{O}(\alpha T H^2)$ for the latter term. Towards invoking Lemma 4 (a variant of Lemma 3.1 adapted from Luo et al. [2021]) to bound $(\star)$, we first aim to bound the following expression using b_t and B_t defined in Equations 7 and 8:

$$\sum_s q^{\pi_*}(s) \sum_{t=1}^T \sum_a (\pi_t(a|s) - \pi^*(a|s)) (Q_t^{\tilde{\pi}_t}(s, a) - B_t(s, a)). \quad (20)$$

To accomplish this, we decompose 20 in a similar fashion to Luo et al. [2021] as follows:

$$\begin{aligned}
& \sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \langle \pi_t(\cdot|s) - \pi^*(\cdot|s), Q_t^{\tilde{\pi}_t}(s, \cdot) - B_t(s, \cdot) \rangle \\
& \leq \underbrace{\sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \langle \pi_t(\cdot|s), Q_t^{\tilde{\pi}_t}(s, \cdot) - \hat{Q}_t(s, \cdot) \rangle}_{\text{BIAS-1}} \\
& \quad + \underbrace{\sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \langle \pi^*(\cdot|s), \hat{Q}_t(s, \cdot) - Q_t^{\tilde{\pi}_t}(s, \cdot) \rangle}_{\text{BIAS-2}} \\
& \quad + \underbrace{\sum_{s \in \mathcal{S}} q^{\pi^*}(s) \sum_{t=1}^T \langle \pi_t(\cdot|s) - \pi_*(\cdot|s), \hat{Q}_t(s, \cdot) - B_t(s, \cdot) \rangle}_{\text{REG-TERM}}.
\end{aligned}$$

Bound on REG-TERM.

First note that by the definition of bonuses in Equations 7 and 8, we have that

$$\begin{aligned}
b_t(s, a) &= \frac{3\gamma H + H(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} \leq 3H + H \leq \mathcal{O}(H), \\
B_t(x, a) &\leq H \left(1 + \frac{1}{H}\right)^H \sup_{s', a'} b_t(s', a') \leq 3H \times \mathcal{O}(H) \leq \mathcal{O}(H^2).
\end{aligned}$$

Since $\eta \leq 1/24H^3$, $\gamma = 2\eta H$, and our definitions of $\hat{Q}_t$ (Equation 6), b_t (Equation 7), and B_t (Equation 8) are parallel to Luo et al. [2021], the bounds from Lemma C.4 of Luo et al. [2021] hold, so namely, we have that $\eta|\hat{Q}_t(s, a) - B_t(s, a)| \leq 1$. This, along with the fact that π_t is generated via exponential weight updates with loss vectors $\hat{Q}_t(s, \cdot) - B_t(s, \cdot)$, means that we can apply the exponential weights bound (Lemma 3) on each state s to obtain

$$\underbrace{\sum_{t=1}^T \sum_{a \in \mathcal{A}} (\pi_t(a|s) - \pi^*(a|s)) \left(\hat{Q}_t(s, a) - B_t(s, a) \right)}_{\text{REG-TERM}} \leq \frac{\log A}{\eta} + \eta \sum_{t=1}^T \sum_{a \in \mathcal{A}} \pi_t(a|s) \left(\hat{Q}_t(s, a) - B_t(s, a) \right)^2.$$

Then, summing over states with weights $q^*(s)$, we get

$$\begin{aligned}
\text{REG-TERM} &\leq \frac{H \log A}{\eta} + \eta \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) \left(\widehat{Q}_t(s, a) - B_t(s, a) \right)^2 \\
&\leq \frac{H \log A}{\eta} + 2\eta \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) \widehat{Q}_t(s, a)^2 \\
&\quad + 2\eta \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) B_t(s, a)^2.
\end{aligned}$$

By Lemma C.3 and C.4 from Luo et al. [2021], we are able to bound the last two terms as such:

$$\begin{aligned}
2\eta \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) \widehat{Q}_t(s, a)^2 &\leq \frac{H}{4\eta} \log \frac{H}{\delta} + \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) \frac{\gamma H}{\bar{q}_t(s, a) + \gamma}, \\
2\eta \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) B_t(s, a)^2 &\leq \frac{1}{H} \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) B_t(s, a).
\end{aligned}$$

Chaining the results together, we have with probability at least $1 - \mathcal{O}(\delta)$,

$$\text{REG-TERM} \leq \tilde{O}\left(\frac{H}{\eta}\right) + \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) \left(\frac{\gamma H}{\bar{q}_t(s, a) + \gamma} + \frac{B_t(s, a)}{H} \right).$$

Bound on BIAS-1 and BIAS-2.

BIAS-1 measures the amount of underestimation that $\widehat{Q}_t$ does of $Q^{\tilde{\pi}_t}$ (related to π_t), while BIAS-2 measures the amount of overestimation that $\widehat{Q}_t$ does of $Q^{\tilde{\pi}_t}$ (related to π^*). Note that $\widehat{Q}_t$ is constructed using $L_{t,h}$ observed from taking $\tilde{\pi}$ and our occupancy measures are with respect to $\tilde{\pi}$. Therefore, using identical proof techniques to Lemmas C.1 and C.2 of Luo et al. [2021], we can bound the terms (with probability at least $1 - \mathcal{O}(\delta)$) as follows:

$$\begin{aligned}
\text{BIAS-1} &\leq \tilde{O}\left(\frac{H}{\eta}\right) + \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) \left(\frac{2\gamma H + H(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} \right), \\
\text{BIAS-2} &\leq \tilde{O}\left(\frac{H}{\eta}\right).
\end{aligned}$$

Applying Lemma 4 (Modified from 3.1 from Luo et al. [2021]).

Combining previous results, we have that with probability at least $1 - \mathcal{O}(\delta)$,

BIAS1 + BIAS2 + REG-TERM

$$\begin{aligned} &\leq \tilde{\mathcal{O}}\left(\frac{H}{\eta}\right) + \sum_{t=1}^T \sum_s \sum_{a \in \mathcal{A}} q^*(s) \pi_t(a|s) \left(\frac{3\gamma H + H(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} + \frac{B_t(s, a)}{H} \right) \\ &\leq \tilde{\mathcal{O}}\left(\frac{H}{\eta}\right) + \sum_{t=1}^T \sum_{s, a} q^*(s) \pi^*(a|s) b_t(s, a) + \frac{1}{H} \sum_{t=1}^T \sum_{s, a} q^*(s) \pi_t(a|s) B_t(s, a), \end{aligned}$$

Thus, we've verified the bound on Equation 20 and can apply Lemma 4 to bound $(\star)$. We also add back in the $\mathcal{O}(\alpha TH^2)$ from before. Then, using 9 (which compares the occupancy measures under $\tilde{\pi}$ and π), we bound $R(T)$ as such (let $\hat{P}_t$ be the optimistic transition function chosen in Equation 8 at episode t)

$$\begin{aligned} R(T) &= \tilde{\mathcal{O}}\left(\frac{H}{\eta} + \sum_{t=1}^T \sum_{s, a} q^{\hat{P}_t, \pi_t}(s, a) b_t(s, a) + \alpha TH^2\right) \\ &= \tilde{\mathcal{O}}\left(\frac{H}{\eta} + \sum_{t=1}^T \sum_{s, a} q^{\hat{P}_t, \pi_t}(s) \pi_t(a|s) \frac{\gamma H + H(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} + \alpha TH^2\right) \\ &\leq \tilde{\mathcal{O}}\left(\frac{H}{\eta} + \frac{1}{(1 - \alpha)^H} \sum_{t=1}^T \sum_{s, a} q^{\hat{P}_t, \tilde{\pi}_t}(s) \pi_t(a|s) \frac{\gamma HA + HA(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{(\bar{q}_t(s) + \gamma)\alpha} + \alpha TH^2\right) \\ &\quad (\text{Lemma 9, and } \tilde{\pi}_t(a|s) \geq \alpha/A) \\ &= \tilde{\mathcal{O}}\left(\frac{H}{\eta} + \frac{1}{(1 - \alpha)^H} \sum_{t=1}^T \sum_{s, a} \left(\frac{HA(\bar{q}_t(s, a) - \underline{q}_t(s, a)) + \eta H^2 A}{\alpha} \right) + \alpha TH^2\right) \\ &\quad (\text{since } q^{\hat{P}_t, \tilde{\pi}_t}(s, a) \leq \bar{q}_t(s, a) \text{ and } \gamma = 2\eta H) \\ &\leq \tilde{\mathcal{O}}\left(\frac{H}{\eta} + \frac{1}{(1 - \alpha)^H} \left(\frac{SH^2 A^{3/2} \sqrt{T} + \eta SA^2 H^2 T}{\alpha} \right) + \alpha TH^2\right) \\ &\leq \tilde{\mathcal{O}}\left(\frac{H}{\eta} + (1 + \alpha H) \left(\frac{SH^2 A^{3/2} \sqrt{T} + \eta SA^2 H^2 T}{\alpha} \right) + \alpha TH^2\right) \\ &\leq \tilde{\mathcal{O}}\left(\frac{H}{\eta} + \frac{SH^2 A^{3/2} \sqrt{T}}{\alpha} + \frac{\eta SA^2 H^2 T}{\alpha} + SH^3 \sqrt{AT} + \eta SAH^3 T + \alpha TH^2\right). \end{aligned}$$

where the fifth line is from Lemma 4 of Jin et al. [2020] and the sixth line follows from noting that for $\alpha \ll 1/2H$ (true for large T),

$$\left(\frac{1}{1 - \alpha}\right)^H \leq (1 + 2\alpha)^H \leq \exp(2\alpha H) \leq 1 + \mathcal{O}(\alpha H).$$

Finally, plugging in $\eta = \min \{1/24H^3, 1/\sqrt{HSAT}\}$ and $\alpha = \mathcal{O}(H^{-1}S^{\frac{1}{2}}A(\log A)^{\frac{1}{3}}T^{-\frac{1}{4}})$ yields with probability at least $1 - \mathcal{O}(\delta)$,

$$\text{Term II} \leq \tilde{O} \left(H^4 + H^3 S A^{\frac{1}{2}} T^{\frac{3}{4}} + H S^{\frac{1}{2}} A T^{\frac{3}{4}} \right).$$

□

C.5 Comparing Occupancy Measures under $\tilde{\pi}$ and π

Lemma 9 (Occupancy Domination) *Let P be a transition kernel, π be a policy, and $\tilde{\pi}$ be the policy π when mixed with α exploration. Then, for all s , $q^{P,\pi}(s) \leq (1 - \alpha)^{-H} q^{P,\tilde{\pi}}(s)$.*

Proof: We first note the pointwise domination, i.e. for all s, a ,

$$(1 - \alpha)\pi(a|s) \leq \tilde{\pi}(a|s) \implies \pi(a|s) \leq \frac{1}{1 - \alpha} \tilde{\pi}(a|s). \quad (21)$$

We prove the claim via induction on h . For $h = 0$, both processes start from the same initial state, hence $q^{P,\pi}(s_0) = q_0^{P,\tilde{\pi}}(s_0)$ for all x . Next, assume

$$q^{P,\pi}(s_h) \leq (1 - \alpha)^{-h} q^{P,\tilde{\pi}}(s_h). \quad (22)$$

holds for any s in layer h , for some $h \in \{0, 1, \dots, H - 1\}$. For any next-layer state s_{h+1} , by the occupancy recursion,

$$\begin{aligned} q^{P,\pi}(s_{h+1}) &= \sum_s \sum_a q^{P,\pi}(s_h) \pi(a|s_h) P(s_{h+1} | s_h, a) \\ &\leq \frac{1}{1 - \alpha} \sum_x q^{P,\pi}(s_h) \sum_a \tilde{\pi}_h(a | x) P(s_{h+1} | s, a) \\ &\quad \text{by (21)} \\ &\leq \frac{1}{(1 - \alpha)^{h+1}} \sum_x \sum_a q_h^{P,\tilde{\pi}}(x) \tilde{\pi}_h(a | x) P(s_{h+1} | s, a) \\ &\quad \text{by the induction hypothesis} \\ &= \frac{1}{(1 - \alpha)^{h+1}} q^{P,\tilde{\pi}}(s_{h+1}). \end{aligned}$$

Thus, we close the induction. The claim follows by noting that $h \leq H$. □

C.6 Bound on Term III

Lemma 10 *With the players following Algorithm 1 with $\alpha = \mathcal{O}(H^{-1}S^{\frac{1}{2}}A(\log A)^{\frac{1}{3}}T^{-\frac{1}{4}})$, we have that with probability at least $1 - 2\delta$,*

$$\underbrace{\sum_{t=1}^T V^{\pi_*, \pi_t^f}(s_0) - V^{\pi_*, \pi_*^f}(s_0)}_{\text{Term III}} \leq \mathcal{O}\left(H^2 S^{\frac{1}{2}} AK T^{\frac{1}{4}} \frac{\log(SAT/\delta)}{\beta} \cdot \left(\frac{\log T}{\varepsilon^2} + \log\left(\frac{SAK}{\delta}\right)\right)\right)$$

Before presenting the proof, we make a few remarks. The term under consideration measures the follower's hypothetical performance when the leader follows the optimal policy. In Yu et al. [2022], an analogous term is used to bound the follower's performance when the leader pulls the optimal action in hindsight. There, this term is controlled by lower-bounding the number of times the optimal arm in hindsight, a^* , is pulled; this is precisely the role of the exploration parameter α .

In our setting, however, the leader will generally never follow the optimal policy π^* . Crucially, the follower's success does not depend on the leader's policy itself, but rather on whether the leader pulls each arm sufficiently often at each state. From the follower's perspective, the interaction induces a latent bandit problem in which the leader's actions and the evolving state together play the role of nature's actions Maillard and Mannor [2014]. Consequently, sufficient exploration across state-action pairs is enough to ensure good follower performance, even when the leader deviates substantially from π^* .

Proof: Because both value functions in Term III use the same leader policy π_* , and because the follower's action does not affect the transition kernel, they induce the same state-leader-action occupancy measure. Therefore,

$$\text{Term III} = \sum_{t=1}^T \left(V^{\pi_*, \pi_t^f}(s_0) - V^{\pi_*, \pi_f^*}(s_0) \right) \quad (23)$$

$$= \sum_{t=1}^T \sum_{h=0}^{H-1} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} q_h^{\pi_*}(s, a) \left[l(s, a, \pi_t^f(s, a)) - l(s, a, \pi_f^*(s, a)) \right] \quad (24)$$

$$\leq \sum_{t=1}^T \sum_{h=0}^{H-1} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} q_h^{\pi_*}(s, a) \mathbb{I}\left\{ \pi_t^f(s, a) \neq \pi_f^*(s, a) \right\}, \quad (25)$$

where the final inequality follows from $l \in [0, 1]$.

We next bound the number of episodes during which the follower uses an incorrect response

at a fixed pair $(s, a) \in \mathcal{S} \times \mathcal{A}$. Let $1 \leq t_1 < t_2 < \dots < t_{n_{(s,a)}(T)} \leq T$ denote the episodes in which the learner's trajectory visits state s and the leader chooses action a . Set $t_0 := 0$ and $t_{n_{(s,a)}(T)+1} := T$. Define the inter-visit lengths $u_i(s, a) := t_i - t_{i-1}$, $i \in [n_{(s,a)}(T) + 1]$, and also define $u_{\max}(s, a) := \max_{i \in [n_{(s,a)}(T)+1]} u_i(s, a)$.

The follower updates its response at (s, a) only when (s, a) is visited. Hence, between two consecutive visits, its response remains unchanged. In particular, for every $t \in (t_{i-1}, t_i]$, $\pi_t^f(s, a) = \pi_{t_i}^f(s, a)$. Consequently,

$$\sum_{t=1}^T \mathbb{I}\left\{\pi_t^f(s, a) \neq \pi_f^*(s, a)\right\} \leq u_{n_{(s,a)}(T)+1}(s, a) + \sum_{i=1}^{n_{(s,a)}(T)} u_i(s, a) \mathbb{I}\left\{\pi_{t_i}^f(s, a) \neq \pi_f^*(s, a)\right\} \quad (26)$$

$$\leq u_{\max}(s, a) \left(1 + \sum_{i=1}^{n_{(s,a)}(T)} \mathbb{I}\left\{\pi_{t_i}^f(s, a) \neq \pi_f^*(s, a)\right\}\right) \quad (27)$$

$$= u_{\max}(s, a) \left(1 + \sum_{k \neq \pi_f^*(s, a)} n_k^f(n_{(s,a)}(T))\right), \quad (28)$$

where we recall that $n_k^f(n_{(s,a)}(T))$ denotes the number of times the follower selects action k during the first $n_{(s,a)}(T)$ visits to (s, a) .

Using $q_h^{\pi^*}(s, a) \leq 1$ in (25), summing over h , and then applying (28), we obtain

$$\text{Term III} \leq H \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \sum_{t=1}^T \mathbb{I}\left\{\pi_t^f(s, a) \neq \pi_f^*(s, a)\right\} \quad (29)$$

$$\leq H \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} u_{\max}(s, a) \left(1 + \sum_{k \neq \pi_f^*(s, a)} n_k^f(n_{(s,a)}(T))\right). \quad (30)$$

We now control the maximum inter-visit length. By Lemma 5, conditional on the history before any episode, the probability of visiting (s, a) is at least $p := \frac{\alpha\beta}{A}$. Hence, for any fixed interval of m consecutive episodes, $\mathbb{P}((s, a) \text{ is not visited during this interval}) \leq (1 - p)^m$. Since there are at most T possible starting episodes for such an interval, a union bound gives

$$\mathbb{P}(u_{\max}(s, a) \geq m) \leq T(1 - p)^m \leq Te^{-pm}.$$

Setting $m = \frac{1}{p} \log \frac{T}{\delta} = \frac{A}{\alpha\beta} \log \frac{T}{\delta}$ thus gives, with probability at least $1 - \delta$,

$$u_{\max}(s, a) \leq \frac{A}{\alpha\beta} \log \frac{T}{\delta}. \quad (31)$$

Finally, union bounding over all $(s, a) \in \mathcal{S} \times \mathcal{A}$ gives, with probability at least $1 - \delta$,

$$u_{\max}(s, a) \leq \frac{A}{\alpha\beta} \log \frac{SAT}{\delta} \quad \text{simultaneously for all } (s, a).$$

By Lemma 6, with probability at least $1 - \delta$, simultaneously for every (s, a) ,

$$\sum_{k \neq \pi_f^*(s, a)} n_k^f(n_{(s, a)}(T)) \leq \mathcal{O}\left(K \left(\frac{\log T}{\varepsilon^2} + \log \frac{SAK}{\delta}\right)\right). \quad (32)$$

Since the right-hand side is at least a constant after adjusting the universal constant, the additive 1 in (30) can be absorbed into this bound. Combining (30), (31), and (32), we obtain, with probability at least $1 - 2\delta$,

$$\text{Term III} \leq \mathcal{O}\left(HSA \frac{A}{\alpha\beta} \log \frac{SAT}{\delta} \cdot K \left(\frac{\log T}{\varepsilon^2} + \log \frac{SAK}{\delta}\right)\right). \quad (33)$$

Finally, under the stated choice $\alpha = \mathcal{O}(H^{-1}S^{1/2}A(\log A)^{1/3}T^{-1/4})$, we have (absorbing logarithmic factors),

$$\text{Term III} \leq \mathcal{O}\left(H^2S^{1/2}AKT^{1/4} \frac{\log(SAT/\delta)}{\beta} \left(\frac{\log T}{\varepsilon^2} + \log \frac{SAK}{\delta}\right)\right). \quad (34)$$

□

C.7 Bound on $R_{\text{Stack}}(T)$

We now are ready to prove Theorem 1 which we restate here:

Theorem 4 *Setting $\alpha = \mathcal{O}(H^{-1}S^{\frac{1}{2}}A(\log A)^{\frac{1}{3}}T^{-\frac{1}{4}})$, Algorithm 1 achieves Stackelberg regret with probability at least $1 - \mathcal{O}(\delta)$:*

$$R_{\text{Stack}}(T) \leq \tilde{\mathcal{O}}\left(\frac{H^2SAKT^{\frac{1}{4}}}{\beta\varepsilon^2} + H^4 + H^3SA^{\frac{1}{2}}T^{\frac{3}{4}} + HS^{\frac{1}{2}}AT^{\frac{3}{4}}\right)$$

where $\tilde{\mathcal{O}}$ hides logarithmic dependencies.

Proof: Suppressing logarithmic dependencies and combining the results in Lemmas 7, 8, and 10 along with the original decomposition, we get that with probability $1 - \mathcal{O}(\delta)$:

$$R_{\text{Stack}}(T) \leq \mathcal{O} \left(SAK \left(\frac{\log T}{\varepsilon^2} + \log \left(\frac{SAK}{\delta} \right) \right) + H \sqrt{T \log \left(\frac{1}{\delta} \right)} \right) \quad (\text{I})$$

$$+ \tilde{\mathcal{O}} \left(H^4 + H^3 S A^{\frac{1}{2}} T^{\frac{3}{4}} + H S^{\frac{1}{2}} A T^{\frac{3}{4}} \right) \quad (\text{II})$$

$$+ \mathcal{O} \left(H^2 S^{\frac{1}{2}} A K T^{\frac{1}{4}} \frac{\log(SAT/\delta)}{\beta} \cdot \left(\frac{\log T}{\varepsilon^2} + \log \left(\frac{SAK}{\delta} \right) \right) \right) \quad (\text{III})$$

$$\leq \tilde{\mathcal{O}} \left(\frac{H^2 S A K T^{\frac{1}{4}}}{\beta \varepsilon^2} + H^4 + H^3 S A^{\frac{1}{2}} T^{\frac{3}{4}} + H S^{\frac{1}{2}} A T^{\frac{3}{4}} \right)$$

□

D Proof of Theorem 2

Proof: Sample $J \sim \text{Unif}([T])$ once before the episode and execute $\tilde{\pi}_J$ throughout. Conditioning on J gives the exact identity

$$V^{\hat{\pi}, \pi_f^*}(s_0) = \frac{1}{T} \sum_{t=1}^T V^{\tilde{\pi}_t, \pi_f^*}(s_0).$$

Hence

$$\begin{aligned} V^{\hat{\pi}, \pi_f^*}(s_0) - V^{\pi_*, \pi_f^*}(s_0) &= \frac{1}{T} \sum_{t=1}^T \left(V^{\tilde{\pi}_t, \pi_f^*}(s_0) - V^{\pi_*, \pi_f^*}(s_0) \right) \\ &= \frac{R_{\text{Stack}}(T)}{T}. \end{aligned}$$

Theorem 1 then gives the stated $\tilde{\mathcal{O}}(T^{-1/4})$ ex-ante guarantee. □

E Proof of Theorem 3

Proof: Fix t and define the local Bellman gap

$$g_t(s) := \sum_{a \in \mathcal{A}} (\tilde{\pi}_t(a | s) - \pi_*(a | s)) Q^{\pi_*, \pi_f^*}(s, a).$$

Because $\pi_\star$ is Bellman-optimal at every state in the induced leader MDP, $g_t(s) \geq 0$. The performance-difference lemma gives

$$\Delta_t(s_0) := V^{\tilde{\pi}_t, \pi_f^\star}(s_0) - V^{\pi_\star, \pi_f^\star}(s_0) = \sum_{h=0}^{H-1} \sum_{s \in \mathcal{S}_h} q_h^{\tilde{\pi}_t}(s) g_t(s).$$

Under Assumption 3, every layer-compatible state has occupancy at least β under every leader policy. Since all summands are nonnegative,

$$g_t(s) \leq \frac{\Delta_t(s_0)}{\beta}.$$

Now use the pointwise Markov average $\bar{\pi}(a | s) = \frac{1}{T} \sum_t \tilde{\pi}_t(a | s)$. Linearity is applied only to the local gap, with the continuation values under $\pi_\star$ fixed:

$$\begin{aligned} g_{\bar{\pi}}(s) &:= \sum_a (\bar{\pi}(a | s) - \pi_\star(a | s)) Q^{\pi_\star, \pi_f^\star}(s, a) \\ &= \frac{1}{T} \sum_{t=1}^T g_t(s) \leq \frac{R_{\text{Stack}}(T)}{\beta T}. \end{aligned}$$

Let $D_h := \max_{s \in \mathcal{S}_h} \{V^{\bar{\pi}, \pi_f^\star}(s) - V^{\pi_\star, \pi_f^\star}(s)\}$. Subtracting the Bellman equations gives, for $s \in \mathcal{S}_h$,

$$\begin{aligned} V^{\bar{\pi}, \pi_f^\star}(s) - V^{\pi_\star, \pi_f^\star}(s) &= g_{\bar{\pi}}(s) + \mathbb{E}_{a \sim \bar{\pi}(\cdot | s), s' \sim P(\cdot | s, a)} [V^{\bar{\pi}, \pi_f^\star}(s') - V^{\pi_\star, \pi_f^\star}(s')] \\ &\leq \frac{R_{\text{Stack}}(T)}{\beta T} + D_{h+1}. \end{aligned}$$

Since $D_H = 0$, backward induction yields

$$D_h \leq \frac{H-h}{\beta} \frac{R_{\text{Stack}}(T)}{T}.$$

Therefore, for every $s \in \mathcal{S}_h$,

$$V^{\bar{\pi}, \pi_f^\star}(s) - V^{\pi_\star, \pi_f^\star}(s) \leq \frac{H-h}{\beta} \frac{R_{\text{Stack}}(T)}{T} \leq \frac{H}{\beta} \frac{R_{\text{Stack}}(T)}{T},$$

which proves the state-wise guarantee. $\square$